



Offering an Evidence-Based Recommendation for Improving the Storm Prediction Center's Categorical Outlook

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ABSTRACT

The Storm Prediction Center (SPC) messages severe thunderstorm hazard likelihood using the categorical outlook, which labels increasing risk through the words “Marginal”, “Slight”, “Enhanced”, “Moderate”, and “High”. Researchers have identified that broadcasters, emergency managers, and members of the public find this word scale is not intuitively ordered and that the five risk words communicate risk inconsistently from person to person. To offer a thoroughly data-driven recommendation on how to more effectively label risk categories in the SPC outlook, we studied the interpretation of four alternative categorical scales by broadcast meteorologists, emergency managers, and members of the public through a mixed-methods approach. Our first study utilizes national survey data collected in the 2023 Severe Weather and Society Survey to test US adults’ risk perception, response efficacy, and likelihood of response when shown the alternative scales. We follow this work with a second study that investigates which alternative broadcast meteorologists and emergency managers (EMs) prefer to use through semi-structured focus groups. Overall, our analysis reveals that a numerical scale leads to more consistent communication outcomes and is generally preferred by partners; however, partners expressed a desire for descriptive words to be attached to those numerical levels, a practice common in the legends of other numerical risk communication scales. The results of both studies suggest that the SPC categorical outlook could communicate risk more clearly through a numerical scale accompanied by a clearly worded legend that uses descriptive words to describe expected impacts for each category.

SIGNIFICANCE STATEMENT

The Storm Prediction Center categorical outlook uses the words “Marginal”, “Slight”, “Enhanced”, “Moderate”, and “High” to communicate severe thunderstorm hazard risk, but these words are unintuitive and could lead to misunderstandings about forecast risk. We test four alternative versions of the categorical outlook scale to see if broadcast meteorologists, emergency managers, and members of the public understand a different set of category labels more easily than the current scale. We find that labeling the categories using numbers is the clearest communication method. Partner interviews also suggest that descriptive words still have

value in categorical risk communication, so we also recommend that a legend using such words be attached to help forecast recipients more quickly learn a number scale.

1. Introduction

As one of the Storm Prediction Center's (SPC) key forecast products, the convective outlook is the primary forecast produced by the National Weather Service (NWS) for tornado, hail, and thunderstorm wind hazards across the Contiguous US for the 1-to-8-day period. Although the outlook is a complex product with multiple components that can be viewed, the convective outlook is most frequently summarized in categorical form. This categorical outlook presents a five-tiered scale of increasing severe thunderstorm hazard risk, and features prominently in the "activity chart" that is the first product displayed on the SPC homepage, in NWS social media and web graphics messaging severe convective hazards (NWS Norman 2024; NWS Lake Charles 2024; NWS Binghamton 2024), and more recently in national television coverage (GMA 2025). Given the key role the outlook plays in severe weather messaging during the lead-up to impactful storms, the outlook has been the focus of recent social science and communication research (Ernst et al. 2021; Williams et al. 2022; Krocak et al. 2022; Trujillo-Falcón et al. 2021; Bitterman et al. 2023a). This research has identified key issues that impact how forecast recipients interpret their risk using the outlook, particularly those surrounding the five words used to indicate increasing levels of risk in the outlook: "Marginal", "Slight", "Enhanced", "Moderate", and "High", in order (see Fig. 1). Past survey experiments and focus group studies have found that these words are unintuitive (Ernst et al. 2021; Williams et al. 2022), both for English speakers and when directly translated into other languages (Trujillo-Falcón et al. 2021; Bitterman et al. 2023a), and that other rank labels such as numbers perform better when communicating increasing severe weather risk (Krocak et al. 2022).

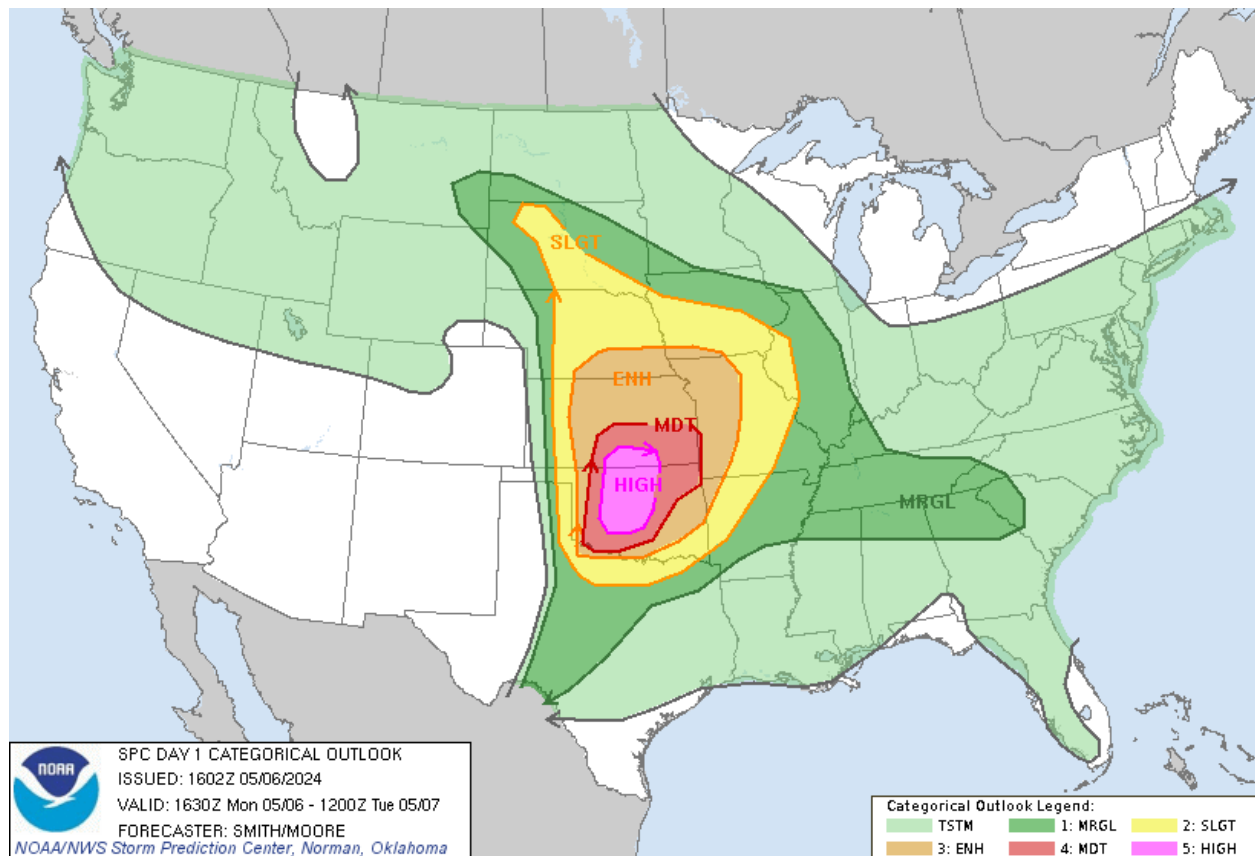


Fig. 1. An example of the Day 1 Categorical Outlook product, the primary risk visual in the SPC convective outlook. The outlook presents 5 named levels of risk: MRGL, or Marginal risk; SLGT, or Slight risk; ENH, or Enhanced risk; MDT, or Moderate risk; and finally HIGH, for High risk of tornadoes, wind, and/or hail. Note the outlook also highlights an area titled TSTM, or General Thunderstorms, in the legend. This area indicates where generic, non-severe thunderstorms are possible on the day of issuance. Image retrieved from <https://www.spc.noaa.gov/exper/archive/event.php?date=20240506>.

Although prior research efforts have highlighted that the current English categorical outlook words impede risk communication, only Bitterman et al. (2023a) has tested a potential replacement for the scale, and only then for the Spanish language version of the outlook. To comprehensively explore and make actionable recommendations on how to improve the English language categorical outlook, we test four alternative word scales: The *Current* scale, which presents the five words currently used in the categorical outlook; the *Likert* scale, which uses words derived from social science survey design theory; the “*Spanish*” scale, which uses English words directly translated from the revised Spanish translation of the *Current* categorical outlook words (as proposed by Trujillo-Falcón et al. 2021 and studied by Bitterman et al. 2023a); and finally the *Numerical* scale, which labels the risk levels as “Level 1 out of 5” and so forth (see

Fig. 2). These scales were presented to members of the public through a national survey, as well as to broadcast meteorologists and emergency managers from across the US through Zoom-mediated focus groups, to gain a better understanding of how different user groups responded to the four options. Using the findings from these experiments as well as past research into reception and interpretation of the categorical outlook scale, we summarized our results by producing recommendations on how to best present increasing risk from severe weather hazards in the SPC categorical outlook.

The Current Scale	The Likert Scale	The “Spanish” Scale	The Numerical Levels
High	Very High	Extreme	Level 5 of 5
Moderate	High	High	Level 4 of 5
Enhanced	Medium	Moderate	Level 3 of 5
Slight	Low	Low	Level 2 of 5
Marginal	Very Low	Minimal	Level 1 of 5

Fig. 2. The four different scales tested in this study, presented by level and color-coded to match the colors used for each level in the Categorical Outlook. The *Current* scale represents the currently used words in the Categorical Outlook, while the *Likert*, “*Spanish*”, and *Numerical* scales represent the three alternative options presented to participants. Note the “*Spanish*” scale is presented in quotations to highlight that it is not presented in Spanish, but instead derived from the words used in the recently updated Spanish Categorical Outlook (see Bitterman et al. 2023a).

2. Background

a. History of the Categorical Outlook

The SPC categorical outlook was first introduced in 1955, and in the decades since has undergone a number of iterative changes. In its first incarnation, the categorical outlook only highlighted areas with a “Moderate” or “High” risk of severe weather, and only on the day that the forecast was issued (Hitchens and Brooks 2012). In 2003, the three levels of the original outlook were tied to a probabilistic outlook of severe weather hazards within 25 miles of a point, allowing for better communication of hazards in addition to easier verification of the outlook forecast using severe weather reports (Ernst et al. 2021). The “Slight” category was then added as a level of forecast risk below “Moderate” in 1974, bringing the total number of categories to three. Few changes were made to the categories until 2014, when the SPC broke the “Slight” category into “Slight” and “Enhanced Slight” categories, the latter of which was shortened to “Enhanced”, while the “Marginal” category was added to better visualize areas of low hazard probabilities that were previously indicated by a label reading “See Text” (Krocak et al. 2021). These words were also translated into Spanish with the help of broadcast meteorologists in 2014, who requested a Spanish version of the outlook that they could present to their viewers (Bitterman et al. 2023a). The expansion of the categorical outlook was intended to give greater resolution to the lower probability areas presented in the probabilistic outlook in the more public-facing categorical outlook (Ernst et al. 2021).

b. Risk Communication with Ordinal Scales

Ordinal scales, like the current categorical outlook, are a common risk communication tool intended to help decisionmakers understand the level of risk they face from a given hazard (e.g. Fujita 1971; WSEC 2006; Batterton and Hale 2017; Burke et al. 2022; EHP 2025; Kastman et al. 2025; NCEI 2025; NDMC 2025; NHC 2025; NWS 2025; SPC 2025; SWPC 2025; VHP 2025). Although ordinal scales are useful for helping relate different categories in an explicit order, the different levels can only be described as greater or less than another, and the intervals between each level of the scale are not necessarily equal. Reducing numerical probabilities to an ordinal scale, as in the categorical outlook, can present risk in a manner perceived as more authoritative and intuitive, but it can lead to neglect of uncertainty and range compression as a continuous scale is converted to a stepwise one (Cox 2008; Krisper 2021).

The communication challenges inherent to ordinal scales can be further exacerbated by the choice of words used to describe each level of the scale, as individuals can interpret words of estimative probability (WEPs) very differently. Multiple studies (MacLeod and Pietraverse 2017; Wintle et al. 2019; Ott 2021; Van Tiel et al. 2022; Hashim 2024; Bodenberger and Thommes 2025) and studies in the meteorological domain (Lenhardt et al. 2020) have identified that when people are asked to associate a number from 0-100 to common WEPs, including “very likely”, “unlikely”, “even chance”, and even words currently used in the categorical outlook like “moderate”, the standard deviation around the mean responses can exceed 15 points while 5-95% percentile ranges often exceed 30 points (MacLeod and Pietraverse 2017; Wintle et al. 2019; Lenhardt et al. 2020; Ott 2021; Van Tiel et al. 2022; Hashim 2024; Bodenberger and Thommes 2025). The WEPs tested in these studies that had the least variation in responses were those that are considered “qualified” (e.g. include a qualifier like “very”, “low”, or “significant” Lenhardt et al. 2020), compared to unqualified WEPs such as “likely”, “possible”, or “expected” (MacLeod and Pietraverse 2017; Lenhardt et al. 2020; Ott 2021; Van Tiel et al. 2022; Hashim 2024). Notably, although most studies only tested the words “Slight”, “Moderate”, and “High” (MacLeod and Pietraverse 2017; Lenhardt et al. 2020; Van Tiel et al. 2022), Williams et al. (2022) tested the words “Enhanced” and “Marginal” along with other WEPs, and found both words were even more broadly interpreted by individuals than the other three categorical levels when asked to assign each a level of severity on a 0-100 scale.

c. Research on Categorical Outlook Communication

As the current five-category, ordinal scale iteration of the categorical outlook has now been operational for over a decade, meteorology and social science researchers have had ample time to study how forecast recipients interpret and use the outlook to message and respond to severe weather hazards. This research has identified that the words used in the categorical outlook are not intuitive to users, and that both experienced partners and the public take issue with the outlook for a variety of reasons (Ernst et al. 2021; Krocak et al. 2021; Williams et al. 2022; Krocak et al. 2022; Bitterman et al. 2023a). Focused research into public and partner interpretation of the outlook began in the late 2010s, where initial work identified how individuals would often misinterpret the order that the risk words are presented in (Ernst et al. 2021; Trujillo-Falcón et al 2021; Williams et al. 2022). This prior work was cited as evidence to

support a series of recommendations for future research into the different components of the convective outlook, including the categorical outlook (Krocak et al. 2021; Bitterman et al. 2023a). Among these recommendations were suggestions to (1) replace the current categorical outlook words with an empirically evaluated alternative, (2) to identify how core partners and the public use the outlook to make decisions and how changes to the outlook could impact those decisions, and (3) to explore the potential of using numerical levels to communicate risk in the categorical outlook (Krocak et al. 2021).

Following the recommendations of Krocak et al. (2021), more recent studies have explored alternative labels for the categorical outlook levels, both in English and Spanish. Krocak et al. (2022) presented members of the public with either one of the risk words “Slight”, “Enhanced”, and “Moderate”; one of the numerical levels “Level 2 of 5”, “Level 3 of 5”, and “Level 4 of 5”; one of the probabilities tied to those levels of 5%, 15%, and 30%; or a combination of the categorical words and the alternate labels, and found that members of the public reported a sequential increase in concern when shown only the numeric labels versus any presentation that included the word labels. Since these findings were published, the SPC has made an effort to include numbers with their categorical outlook products (see legend in Fig. 1), NWS forecast offices have added levels to their products (NWS New Orleans 2024), and broadcast meteorologists have begun to present the outlook with only numerical levels (Forbes 2023; Rugger 2023; Osgood 2025). Researchers have also studied alternative versions of the categorical outlook scale in Spanish, after it was found that Spanish speakers in the U.S. reported that the third level word in the original Spanish outlook levels, “Elevado” (a direct translation of “Enhanced”), and “Alto” (a direct translation of “High”) communicate almost the same amount of risk (Bitterman et al. 2023a). Bitterman et al. (2023a) found that an alternative scale developed by a team of linguists at Penn State University and published by Trujillo-Falcón et al. (2021) that uses the words “Minimo”, “Bajo”, “Moderado”, “Alto”, and “Extremo” was more intuitive and easier to order for Spanish speakers than the direct translations. While different from the English categories, these new translations have since been implemented by the SPC and used in Spanish language versions of the outlook.

d. Scales in risk and hazard communication

Although some possible alternatives to the *Current* categorical outlook risk words have been developed in these outlook-focused studies, consideration of risk communication scales applied to other weather hazards could suggest further alternative designs for the categorical outlook. Most hazard communication scales in the meteorological context present between four and six levels but vary in terms of their time of issuance. Some scales are issued and interpreted in advance of a hazard as a forecast (including the categorical outlook for convection, Excessive Rainfall Outlook or ERO for flooding, the Winter Storm Severity Index or WSSI for winter hazards, and the Space Weather Scales for geomagnetic storms), while others are assessed and assigned during or shortly after an event to define its severity (including the Enhanced Fujita or EF scale for tornadoes, the Saffir-Simpson hurricane wind scale for tropical storm wind speeds, and the NorthEast Snowfall Impact Scale or NESIS for winter storms).

Table 1: List of risk communication scales referenced for this study.

Scale Title	Hazard	Scale Type	Scale Levels (High to Low)
Excessive Rainfall Outlook (ERO)	Rainfall exceeding Flash Flooding Guidance	4-level categorical risk forecast	High Moderate Slight Marginal
Winter Storm Severity Index	Winter Weather Hazards	4-level categorical risk forecast	Extreme Major Moderate Minor
Fire Weather Outlook	Fire Weather Conditions	3-level categorical (with two additional for thunderstorm hazards) risk forecast	Extremely Critical Critical Elevated (Isolated and Scattered Dry Thunderstorms)
Impact-Based Warning Words (Ripberger et al. 2015)	Potential Tornado Damage	6-level categorical hazard forecast	Incredible Devastating Severe Significant Moderate Light
(Enhanced) Fujita Scale	Tornado Damage	6-level numerical impact assessment scale with categorical legend	EF-5 "Incredible" EF-4 "Devastating" EF-3 "Severe" EF-2 "Significant" EF-1 "Moderate" EF-0 "Light"

Saffir-Simpson Hurricane Wind Scale	Hurricane Wind Speed	5-level numerical impact assessment with categorical legend	Category 5 “Catastrophic” Category 4 “Catastrophic” Category 3 “Devastating” Category 2 Category 1
NorthEast Snowfall Impact Scale (NESIS)	Snowfall for given population	5-level numerical impact assessment scale with categorical legend	Category 5 “Extreme” Category 4 “Crippling” Category 3 “Major” Category 2 “Significant” Category 1 “Notable”
HeatRisk Scale	Excessive Heat	5-level numerical risk forecast scale with categorical legend	Level 4 “Extreme” Level 3 “Major” Level 2 “Moderate” Level 1 “Minor” Level 0 “Little to No Risk”
Drought Monitor	Drought	5-level numerical impact assessment scale with categorical legend	D4 “Exceptional” D3 “Extreme” D2 “Severe” D1 “Moderate” D0 “Abnormally Dry”
Modified Mercalli Intensity Scale	Earthquake Shaking	10-level numerical impact assessment scale with categorical legend	X “Extreme” IX “Violent” VIII “Severe” VII “Very Strong” VI “Strong” V “Moderate” IV “Light” III “Weak” II “Weak” I “Not Felt”
Volcano Aviation Color Code	Aviation Volcano Hazards	4-level categorical risk forecast	Red Orange Yellow Green
Volcano Alert Levels	Ground-level Volcano Hazards	4-level categorical risk forecast	Warning Watch Advisory Normal
Space Weather Scales	Solar Radiation Storms, Radio Blackouts, and Geomagnetic Storms	5-level numerical scale impact forecast with categorical legend	5 “Extreme” 4 “Severe” 3 “Strong” 2 “Moderate” 1 “Minor”

While a few of these weather risk scales use words that are similar to the categorical outlook, others communicate categorical hazards using more discipline-specific language. The

Weather Prediction Center's ERO product was developed with the categorical outlook as a model (Burke et al. 2022) and forecasts the probability of exceeding flash flood guidance within 25 miles of a point. Both outlooks have similar scales, though the ERO drops the "Enhanced" category label and only uses "Marginal", "Slight", "Moderate", and "High" for its risk categories (Burke et al. 2023). Alternatively, the WSSI takes a unique approach that ranks forecast winter weather impacts from "Minor", to "Moderate", then "Major", and finally "Extreme" (Kastman et al. 2025). The SPC's Fire Weather Outlook uses even more specialized language, ranking fire weather conditions from "Elevated", to "Critical", and finally "Extremely Critical", while also highlighting areas where Isolated and Scattered "dry" thunderstorms can be anticipated (SPC 2025). The NWS has also made an effort to communicate the significance of tornado hazards in the text that accompanies tornado warnings, which they call Impact-Based Warning (IBW) information (Ripberger et al. 2015). While these consequence-based messages use descriptive words including "Significant" or "Incredible damage" and include protective actions such as seeking shelter or leaving an area, the words "Significant", "Severe", "Devastating", and "Incredible" all elicited similar response rates for likelihood of taking protective action. Further, the rate at which survey respondents reported that they would attempt to evacuate and flee from the tornado increased across these four descriptive words while the rate respondents suggested they would shelter in place decreased, suggesting that overly-aggressive word use could reduce individuals' response efficacy (belief that response actions will effectively protect you) for recommended sheltering actions (Ripberger et al. 2015).

Other scales present their category levels using numbers, although legends that describe these scales use descriptive words for each level to communicate impacts. Foremost among these are the Fujita and Enhanced Fujita (EF) scales that were developed to rate the intensity of tornadoes on a 0-5 scale based on wind speeds and damage (Fujita 1971; WSEC 2006). The impact words tested in tornado warning messages by Ripberger et al. (2015) were pulled directly from this scale, and range from "Light" for EF0 tornadoes, to "Moderate", "Significant", "Severe", "Devastating", and finally "Incredible" for EF5 tornadoes. In the tropical domain, hurricanes are rated on the Saffir-Simpson scale from "Category 1" to "5", again based on maximum windspeeds (NHC 2025). Hurricanes of "Category 3" strength or greater are given the additional moniker "Major" hurricanes, and their damage is described as ranging from "Devastating" to "Catastrophic". Similarly, the NESIS rates snowfall events on a "Category 1"

to “5” scale, although this scale is defined using a calculation of impact that combines snowfall totals with population impacted by snowfall (NCEI 2025). The five levels of the scale are also described using words, which range from “Notable” to “Significant”, “Major”, “Crippling”, and “Extreme”. NWS’ experimental “HeatRisk” scale also presents a number scale with words used to describe each level of the scale, from “Little to No Risk” at level 0 to “Minor”, “Moderate”, “Major”, and “Extreme” at levels 1 through 4 (NWS 2025). Drought hazards are communicated in a 0-4 numerical scale as well, which uses the words “Abnormally Dry”, “Moderate”, “Severe”, “Extreme”, and “Exceptional” to highlight increasing levels of dryness (NDMC 2025).

Other hazards outside of meteorology are also communicated using a variety of scales, which although unique share some foundational similarities. Earthquake shaking intensity, unlike the meteorology scales, is rated on the Modified Mercalli Intensity (MMI) scale from 1 to 10 using roman numerals, with a list of descriptive words for each level of shaking from “not felt” to “weak”, “light”, “moderate”, “strong”, “very strong”, “severe”, “violent”, and “extreme”, with “weak” used for both the second and third level (EHP 2025). In contrast, volcano hazards are communicated using an aviation color code combined with an overlapping alert level similar to the current NWS warning system, with risk increasing from green, to yellow, then orange and red, while the ground activity of the volcano is captured by alert levels that rise from an “advisory” to a “watch” and then “warning” level when an eruption becomes imminent (VHP 2025). Finally, the Space Weather Prediction Center uses three scales that range from “1” to “5” that detail the severity of forecast and ongoing solar radiation storms, radio blackouts, and geomagnetic storms, which use the words “Minor”, “Moderate”, “Strong”, “Severe”, and “Extreme” to describe the impacts of these solar phenomena that can be expected with each numerical level on the scale (SWPC 2025).

Across these varied risk scales, however, only the WSSI, Saffir-Simpson scale, drought monitor, and Volcano Aviation Color Code have seen social science reviews similar to those performed with the categorical outlook (Ernst et al. 2021; Krocak et al. 2021; Williams et al. 2022; Krocak et al. 2022; Bitterman et al. 2023a). First, when surveyed about drought identification tools, decision-makers across North America reported that the drought monitor was the most effective index to use for identifying drought and dry regions when compared to other soil water and vegetation indices (Heim et al. 2023), while decision-makers in North Carolina that participated in focus groups and surveys expressed a need for more context on drought

impacts in the scale (Ward et al. 2022). Study of the Saffir-Simpson hurricane wind scale, on the other hand, have found that the scale is often interpreted as describing the overall risk posed by a tropical storm, when it only expresses the maximum wind in the core of the storm and not the storm surge and flooding hazards that often pose the greatest threat to life and property in these storms and thus should be replaced or supplemented by additional hazard specific scales (Camelo and Mayo 2021; Paxton et al. 2024; Mol et al. 2025). More detailed studies of user preferences that inform scale design have been performed for the Volcano Aviation Color Code and its international permutations (Fearnley et al. 2012; Potter et al. 2014; Fearnley and Beaven 2018), which have found that the scale should not be standardized and instead provide a framework that can be scaled to different stakeholders and their unique contexts. Similarly, the WSSI has been continuously developed with user feedback that has led to fundamental design changes that have shaped the forecast product to meet the expressed needs of its users and the public at large (Semmens et al. 2023, 2024; Kastman et al. 2025). Combined, these findings suggest that risk communication scales are most effective when they are designed and improved with user feedback, versus when message producers create a scale based on the qualities of the hazard and then expect decision-makers to adopt the scale.

e. Alternatives to the Current Categorical Outlook Scale

Taking into account prior categorical outlook research that highlighted the need for a new set of risk labels, the labels used in other operational verbal risk scales, and the knowledge that the most effective risk scales use decision-maker feedback to inform their design, in this study we compared three alternate outlook scales to the *Current* outlook scale (see Fig. 2). The first of those alternatives was a scale based on the five-point Likert scale, which was originally developed for use in psychology research. In survey applications, Likert scales ask respondents to report their level of agreement with a statement by using a scale, the most common version of which uses the words “Strongly Disagree”, “Disagree”, “Neutral”, “Agree”, and “Strongly Agree” (Edmonson 2025; Batterton and Hale 2017). The main advantage of using this scale to communicate risk is that it is ordinal in nature, as it presents a clear rank order unlike the nominal *Current* scale (Batterton and Hale 2017). Further, the Likert scale takes advantage of qualifying words, which studies have shown reduce variation in probabilistic interpretations of WEPs (MacLeod and Pietraville 2017; Lenhardt et al. 2020; Van Tiel et al. 2022; Hashim 2024).

To adapt the Likert scale to the categorical outlook, we replaced the level of agreement with “Low”, “Medium”, and “High”, creating a scale that ranges from “Very Low”, to “Low”, “Medium”, “High”, and finally “Very High”. Throughout this paper, we refer to this alternative scale as the *Likert* scale.

The second alternative scale, the “*Spanish*” scale, is derived from direct English translations of the revised Spanish language version of the outlook (hence the quotations used around “Spanish” when the scale is mentioned). The revised Spanish words “Minimo”, “Bajo”, “Moderado”, “Alto”, and “Extremo”, are currently operational as part of the Spanish version of the categorical outlook (Trujillo-Falcón et al. 2021; Bitterman et al. 2023a). Directly translated back to English, these words create a scale that ranges from “Minimal”, to “Low”, “Moderate”, “High”, and finally “Extreme”. While this scale performs well among Spanish speakers (Bitterman et al. 2023a), its direct translation had not been previously tested among English-speaking members of the public. Its main potential advantage over the *Current* scale is the removal of the categories “Slight” and “Enhanced”, which are cited as the most often mis-ordered words (Ernst et al. 2021; Williams et al. 2022) and have not been tested as WEPs for communicating risk information. The “*Spanish*” scale also uses words that are similar to multiple operational risk scales, including the MMI for earthquakes, the NESIS for winter weather, and the geomagnetic storm scale. Throughout this paper, we refer to this alternative scale as the “*Spanish*” scale.

The final alternative scale we tested is the *Numerical* scale. Initially tested by Krocak et al. (2022), this scale replaces the use of risk words with simple numbers associated with a level, such as “Level 1 of 5”. As noted by our review of risk scales, number-based scales are commonly used to communicate increasing levels of risk for various hazards, both for weather hazards or otherwise (Fujita 1971; WSEC 2006; Theisen 2023; NHC 2025). Similar to the *Likert* scale, the advantage posed by a *Numerical* scale is that it presents an ordinal ranking where a clear order can be interpreted from the scale, even if the difference between each level is not clearly defined (Batterton and Hale 2017). Unlike the *Likert* and “*Spanish*” scales, however, the *Numerical* scale is more easily ordered intuitively, reducing ordering errors that could emerge from the broad interpretations of the WEPs used by the other three scales tested here. While previous tests of the abbreviated *Numerical* scale in the categorical outlook suggested that ordering errors were indeed reduced, the scale was not tested in full, and definitive conclusions

cannot be drawn without knowing how end users will react compared to the *Current* scale (Krocak et al. 2022). Regardless, the *Numerical* scale has become more widely used to communicate severe convective weather risk, with some meteorologists already adopting it for use on television and social media (Capps 2025).

f. Hypotheses and Research Questions

The overarching goal of the experiments we performed with members of the public, emergency managers, and broadcast meteorologists is to explore the question:

When presented with alternative risk communication scales for use in the SPC categorical outlook, which scales are preferred by users, and which scales result in consistent increases in average concern and likelihood of response without impacting response efficacy?

With this goal in mind, we applied the primary findings from Krocak et al. (2021) and Ripberger et al. (2015) to develop two hypotheses:

H1: Average perceived concern and likelihood of protective action will not increase linearly when participants are shown the *Current* SPC scale, while the *Numerical* scale will see the most consistent linear increase in both measures across increasing levels of the scale.

H2: Response efficacy will be consistent across the *Numerical* scale—but will decrease for the highest levels of the word scales (“High”/“Very High”/“Extreme”).

For our second study we were interested in learning emergency manager and broadcaster preferences for each of the four scale options through qualitative and quantitative analysis of focus group participants’ perceptions of each. Given the exploratory nature of this study, instead of hypotheses, we pose a broader research question:

RQ1: What are emergency manager and broadcast meteorologist preferences for the proposed word scales for the categorical outlook?

3. Study 1: Survey of Public Interpretation

a. Methods

Our first goal was to assess public interpretation of the four SPC word scales (see Fig. 2) using a national survey of US adults. Data for this experiment were collected as part of the 2023 Severe Weather and Society survey (WX23), which was designed and fielded by the University of Oklahoma's Institute for Public Policy Research and Analysis (Bitterman et al. 2023b). The survey collected responses from an opt-in web panel maintained by Qualtrics, with a sample size of 1,513 US adults. Qualtrics utilized a dynamic sampling method to solicit participants, whereby they send out a first round of invitations to a sample that matches the demographics of the US population for gender, age, race, ethnicity, and NWS region. As participants completed the survey, Qualtrics sent out subsequent rounds of invitations based on which demographic group quotas had not yet been met. However, as an additional measure to enhance sample representativeness, we also calculated post-stratification survey weights for each respondent using an iterative proportional fitting (raking) procedure. This approach adjusted the sample to align with U.S. Census population benchmarks on age, education, gender, income, race, and National Weather Service (NWS) region (see Table A1 in Appendix A for population margins). Rather than constructing fully cross-classified demographic cells, the raking procedure iteratively aligned the marginal distributions of each characteristic with their corresponding population targets. Respondents who were overrepresented relative to Census benchmarks were assigned weights less than one, whereas those who were underrepresented were assigned weights greater than one. For transparency, we also conducted parallel analyses using the original unweighted data. These results, which are highly comparable to the weighted results presented here, are reported in Appendix B (see Figs. B1a-c, B2a-c).

For the experiment detailed in this study, WX23 respondents were randomly assigned to one of the four scales, and then to a single word from that scale, resulting in 20 randomly assigned groups. An average of 75.65 participants were assigned to each of the 20 groups in the

survey, which ranged in size from 61 to 94 participants. For the survey experiment, respondents were first shown the following text:

Next, we have some questions about a hypothetical upcoming severe weather event. Now imagine that it is a Saturday morning at 8:00 AM and you see this notification: The Storm Prediction Center has issued a [RANDOMIZED WORD] risk of severe weather for your area today.

Wherein the bracketed text indicates that the respondent was shown the single word associated with the randomized scale and level they were assigned (e.g. “Slight” from the *Current* scale). Then respondents were asked to rank their concern level, likelihood of changing plans, and confidence in the actions they would take via the following questions:

On a scale from 0 to 100, where 0 means not at all concerned and 100 means extremely concerned, how concerned would you be if you were to get this forecast?

On a scale from 0 to 100, where 0 means not at all likely and 100 means extremely likely, how likely is it that you would change your plans for the day if you were to get this forecast?

On a scale from 0 to 100, where 0 means not at all confident and 100 means extremely confident, how confident are you that the actions you might take to protect yourself from the severe weather would be effective?

These three questions were chosen to match the survey questions previously issued in Krocak et al. (2022). After answering the above three questions with only the text description of the forecast, respondents were then shown the same text but with the addition of a map indicating their location and a hypothetical SPC convective outlook that corresponded with the scale and level they were randomly assigned (see Fig. 3a-f). After viewing the text below, respondents

were asked the same three questions listed previously about their concern, likelihood of response, and confidence in the actions they might take.

In addition to words and phrases, forecasts often include maps to help depict the risk of severe weather at a given location. Imagine that it is once again Saturday morning at 8:00 AM and you see this forecast and map: The Storm Prediction Center has issued a [RANDOMIZED WORD] risk of severe weather for your area today.”

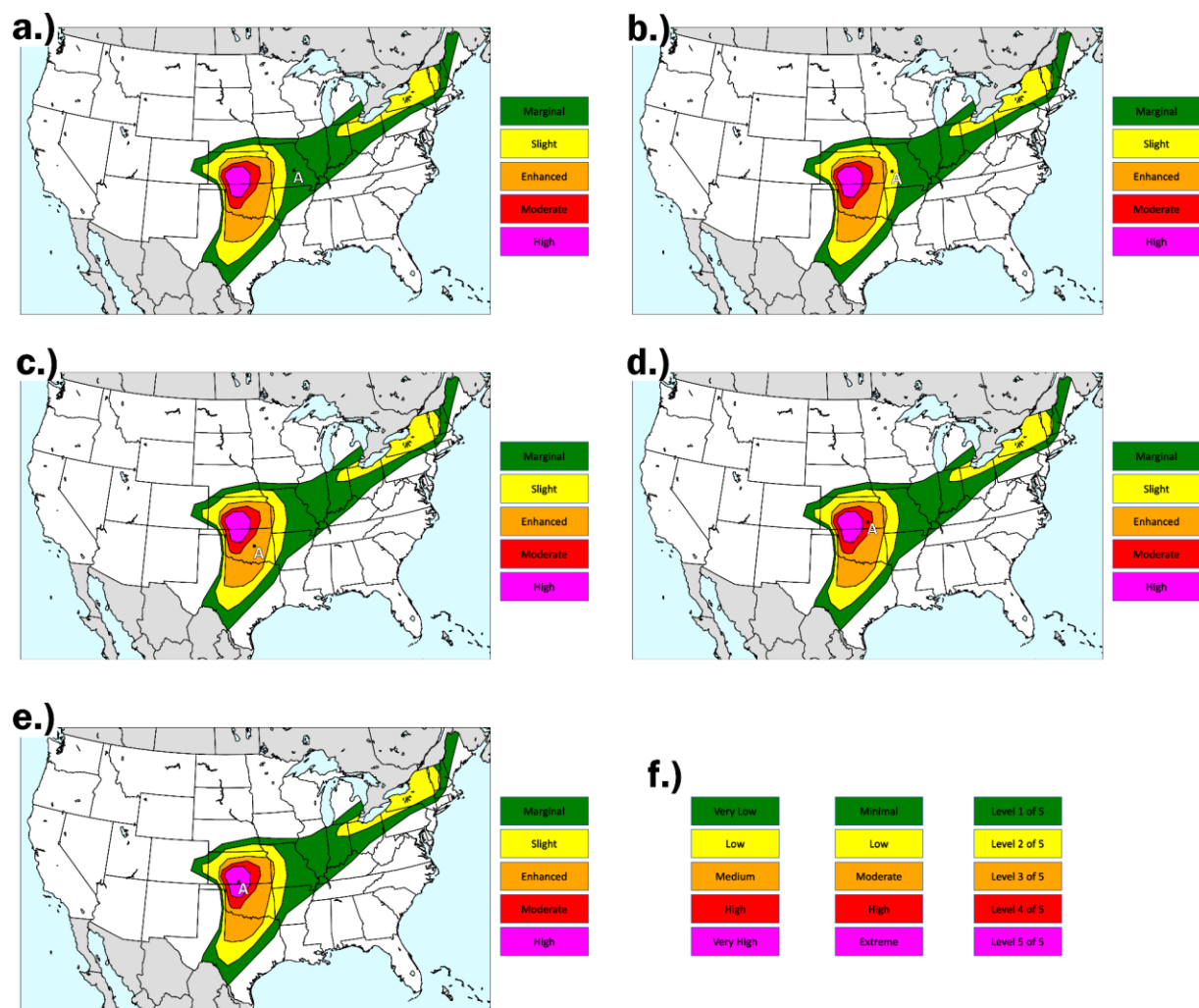


Fig. 3a-f. The five maps WX23 respondents could have been shown if they were assigned to the current SPC scale. Note that participants only saw one map; the one that corresponded to their assigned scale and level. Participants were also told that they are located at point “A”,

which is labeled on the maps for each individual category level. Panel a.) shows the map for the *Current* scale at the first level, panel b.) shows the map for the *Current* scale at the second level, and so on. Panel f.) shows the legends presented for the *Likert*, “*Spanish*”, and *Numerical* scales, which were presented using the same five maps.

Results from the survey questions were compared across levels and across scales, and then differences were assessed between responses to the text descriptions and those to the descriptions that were accompanied by the maps.

b. Results

First, comparing the mean level of concern reported by participants for each of the five words in each scale reveals that concern generally increased across increasing levels of each scale, except for the *Current* scale (see Fig. 4). The *Numerical* scale displayed the most consistent increase in mean concern across its five levels, while participants shown the *Likert* and “*Spanish*” scales rated their concern for the top two and bottom two risk words in each respective scale similarly. Mean concern increased more across the middle three words in the *Likert* and “*Spanish*” scales. Finally, mean responses for the *Current* scale were higher for the first and second level word “Marginal” than the second level word “Slight”, as well as for the third and fourth level words “Enhanced” and “Moderate”, resulting in a “W” pattern for mean concern across the five words in this scale (see Fig. 4).

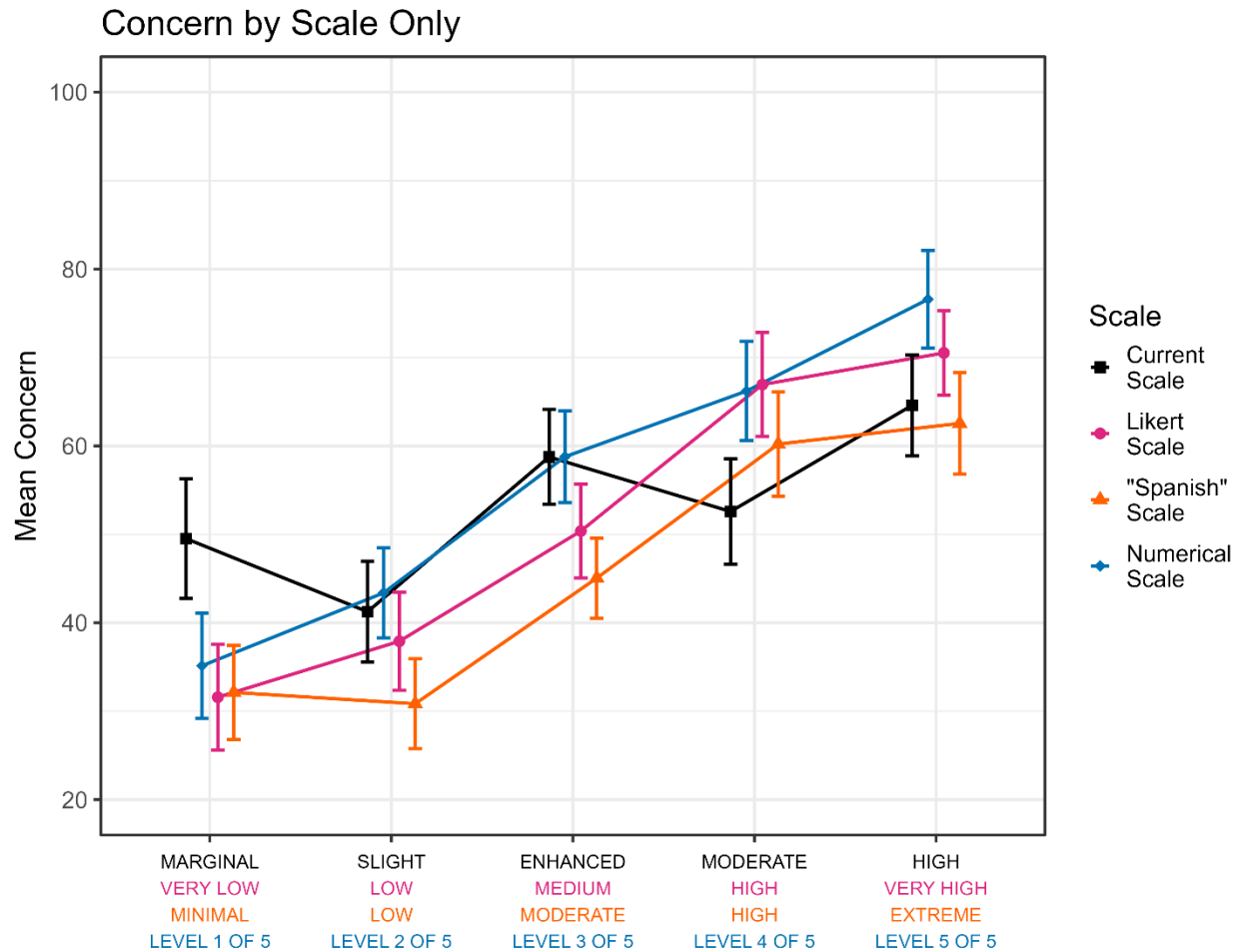


Fig. 4. Weighted mean responses for concern on a scale from 0-100 for participants that were shown only one of the words for each categorical outlook scale. Labels on the X-axis are color coded to match the scale that they are from. Error bars around each mean represent the 95% confidence interval around each, based on the standard error.

When asked to rate the likelihood of changing their plans based on the severe weather forecast, participants displayed very similar response behaviors to their responses for the concern question. As for mean concern, mean likelihood of response increased most consistently across the five levels for the *Numerical* scale, while the “W” pattern observed in mean concern was also observed for mean likelihood of response for the *Current* scale (see Fig. 5). No significant change, and even a slight decrease, in mean likelihood of response was observed between the *Likert* scale words “Very Low” and “Low”, as well as “High” and “Very High”. A more consistent increase in mean likelihood of response was noted for participants shown the “*Spanish*” scale, although there was only a small increase in mean likelihood of response between the words “High” and “Extreme”.

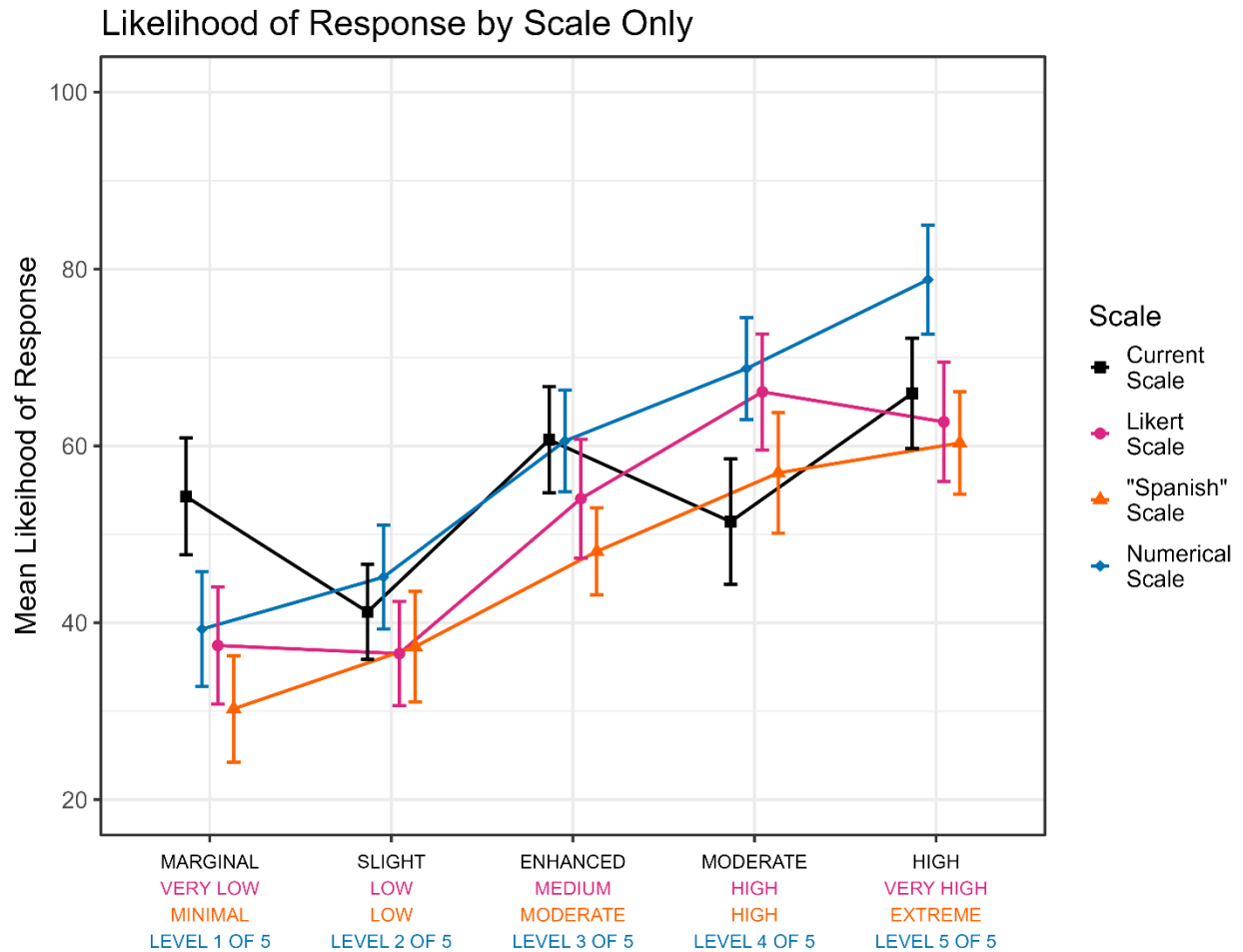


Fig. 5. Same as Fig. 4, but for weighted mean responses for likelihood of response.

When asked whether they felt confident that they could take effective protective actions against the severe weather hazards communicated by the risk scale words, participants' mean responses remained relatively stable across the scales, aside from the *Current* and "*Spanish*" scale (see Fig. 6). Confidence decreased between the *Current* scale words "Marginal" and "Slight" and the "*Spanish*" scale words "Low" and "Moderate", but these inconsistencies did not appear to suggest any pattern in participant interpretations. Confidence in protective action effectiveness increased towards the highest levels of the scale for the *Likert* and "*Spanish*" scales and changed little across the levels of the *Numerical* scale.

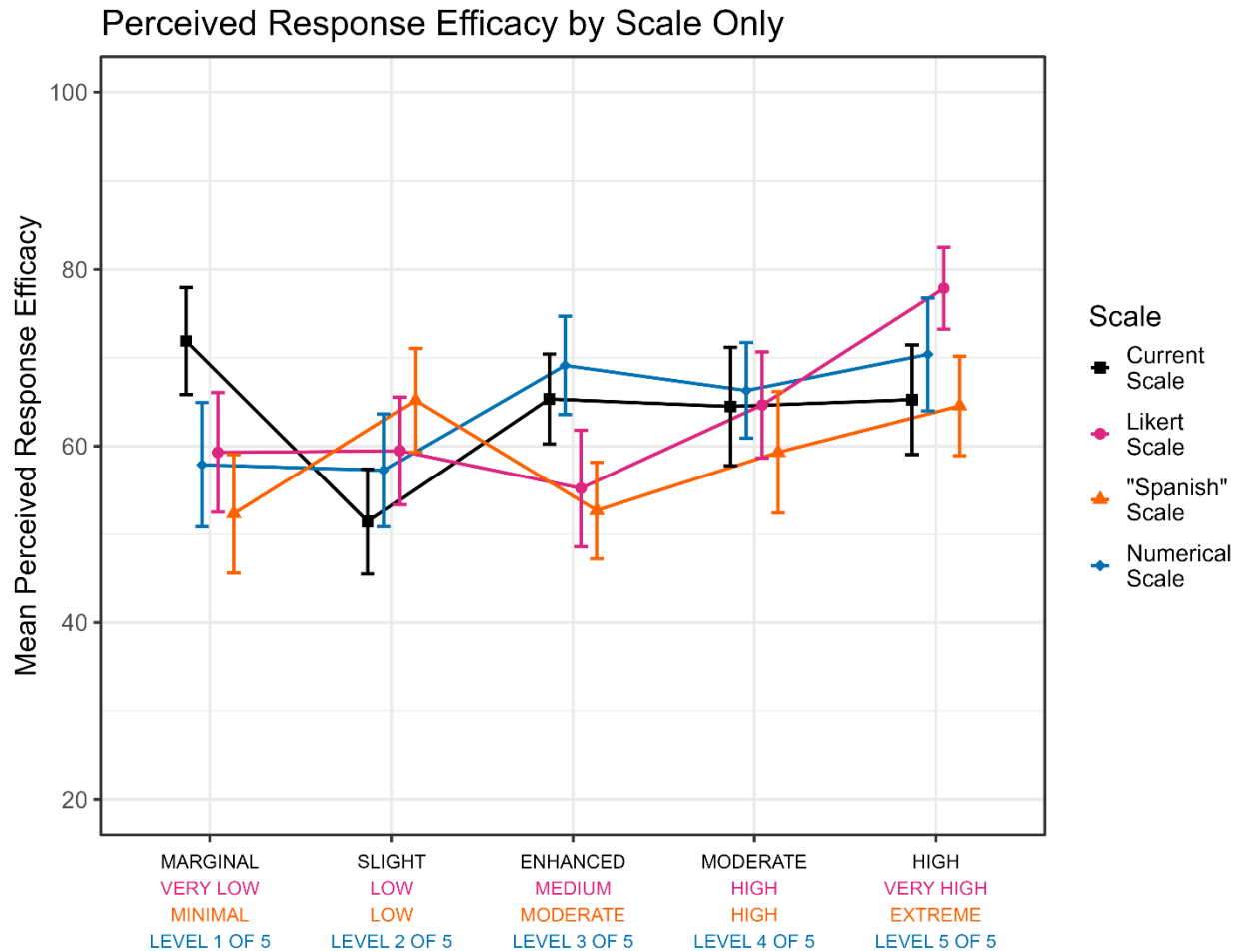


Fig. 6. Same as Fig. 4, but for mean responses for perceived response efficacy.

When participants were shown their location using a categorical outlook map along with a legend for the scale based on the word that they were shown (see Figs. 3a-f), mean responses for concern and likelihood of changing plans became relatively linear across all four scales, with some minor exceptions. Similar levels of average concern were noted between the words “Marginal” and “Slight” for the *Current* scale, “Minimal” and “Low” for the “*Spanish*” scale, and “High” and “Very High” for the *Likert* scale (see Fig. 7). This overall pattern of responses mostly repeated for the response question, although the increases in likelihood of changing plans were more linear across all scales save for at the low end of the *Current* scale and the high end of the *Likert* scale (see Fig. 8). Interestingly, confidence in ability to take action increased from around 55 to around 70 from the lowest levels of the scales to the highest levels, although confidence decreased between the first two levels of the *Current* scale (see Fig. 9). Note that the graphics shown to participants included legends with all category labels presented in rank order

(see Fig. 3f), which likely impacted participants' interpretation of their concern and likelihood of response.

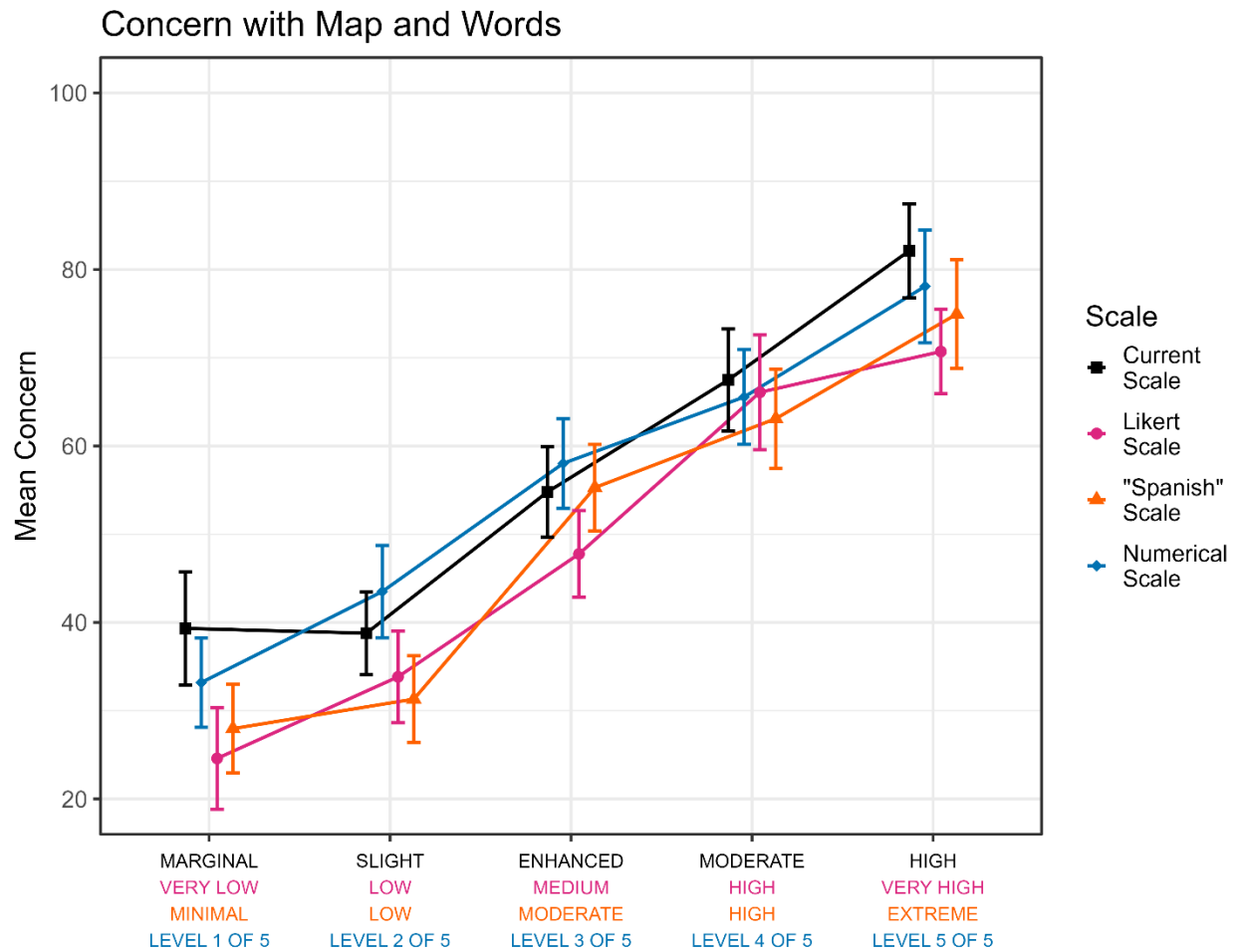


Fig. 7. Mean responses for concern on a scale from 0-100 for participants shown their location on a map of a categorical outlook with an accompanying legend. Labels on the X-axis are color coded to match the scale that they are from. Error bars around each mean represent the 95% confidence interval around each, based on the standard error.

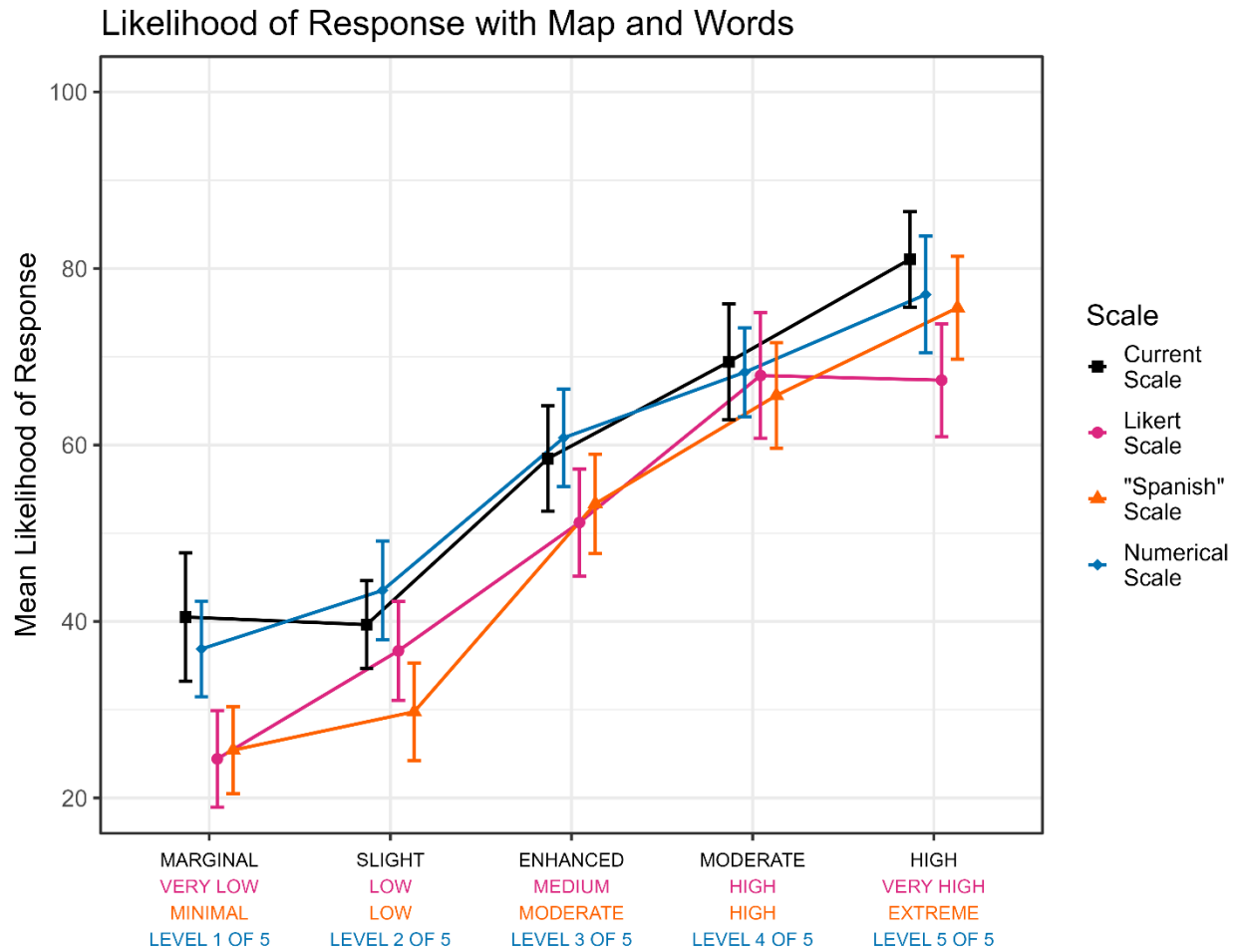


Fig. 8. Same as Fig. 7, but for mean likelihood of response.

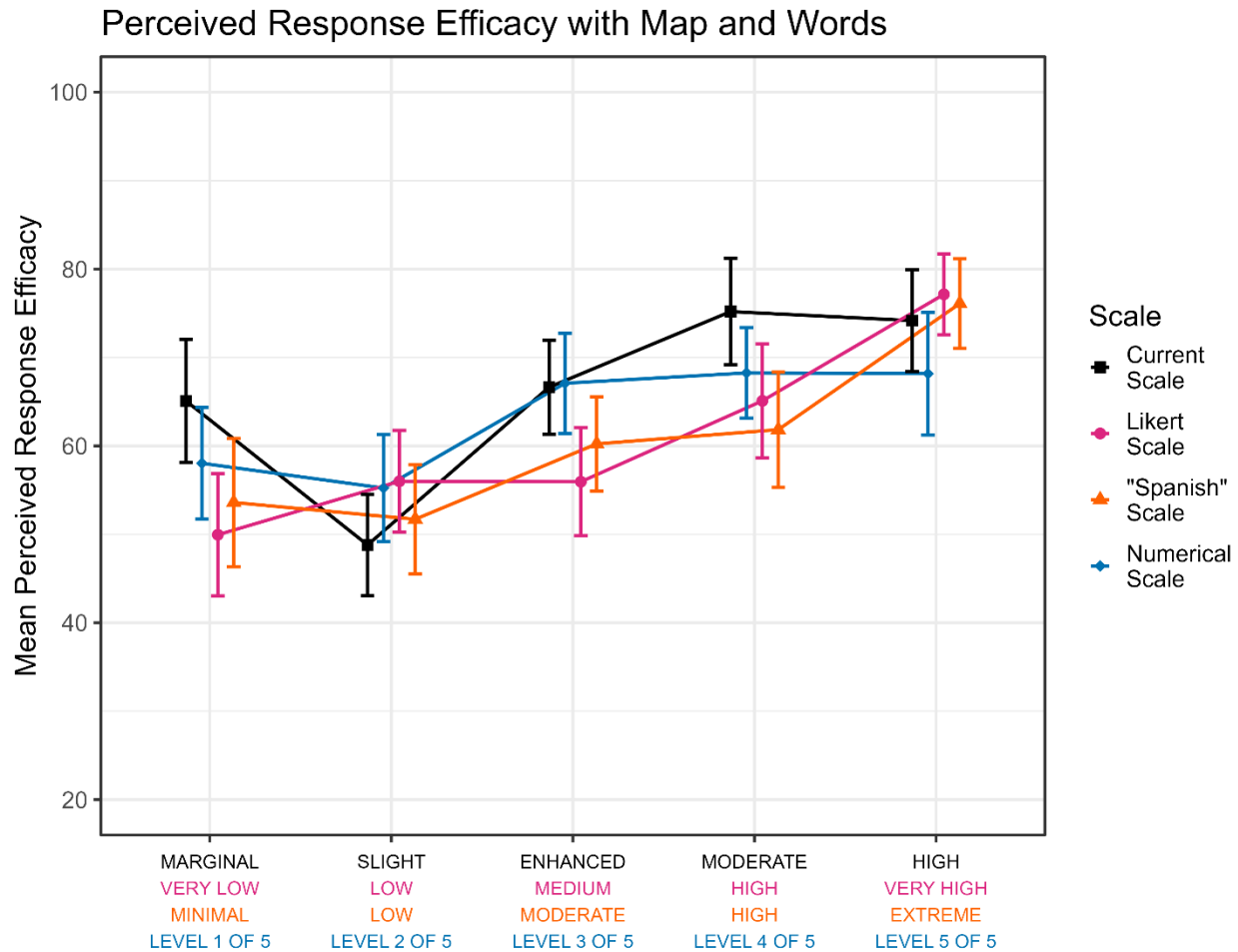


Fig. 9. Same as Fig. 7, but for mean perceived response efficacy.

4. Study 2: Interviews with Emergency Managers and Broadcasters

a. Methods

While our first study focused on how members of the public react to categorical outlook information, in our second study, we complemented this analysis with data collected from emergency managers (EMs) and broadcast meteorologists that receive and make decisions using the outlook regularly. To collect EM and broadcaster responses to our four alternative categorical outlook scales, we ran a series of focus groups as part of the 2024 Hazardous Weather Testbed (HWT) Convective Outlook Innovations Experiment. We invited 88 broadcast meteorologists and EMs to participate in online focus groups across six days in April 2024, with a 50% response rate of 21 broadcasters and 23 EMs that attended (see Fig. 10 for participants' approximate locations of origin). Each of the six focus groups consisted of several morning sessions, where participants were separated into groups based on their status as a broadcast meteorologist or an

EM. The one exception to this separation was the first focus group, where the two groups were merged as only one EM attended. In their focus groups, participants were presented with and then discussed several innovations related to the SPC convective outlook. These morning sessions were followed by an early afternoon session where all broadcaster and EM participants were brought together and asked to share their thoughts on the outlook innovations.

2024 HWT Focus Group Participants

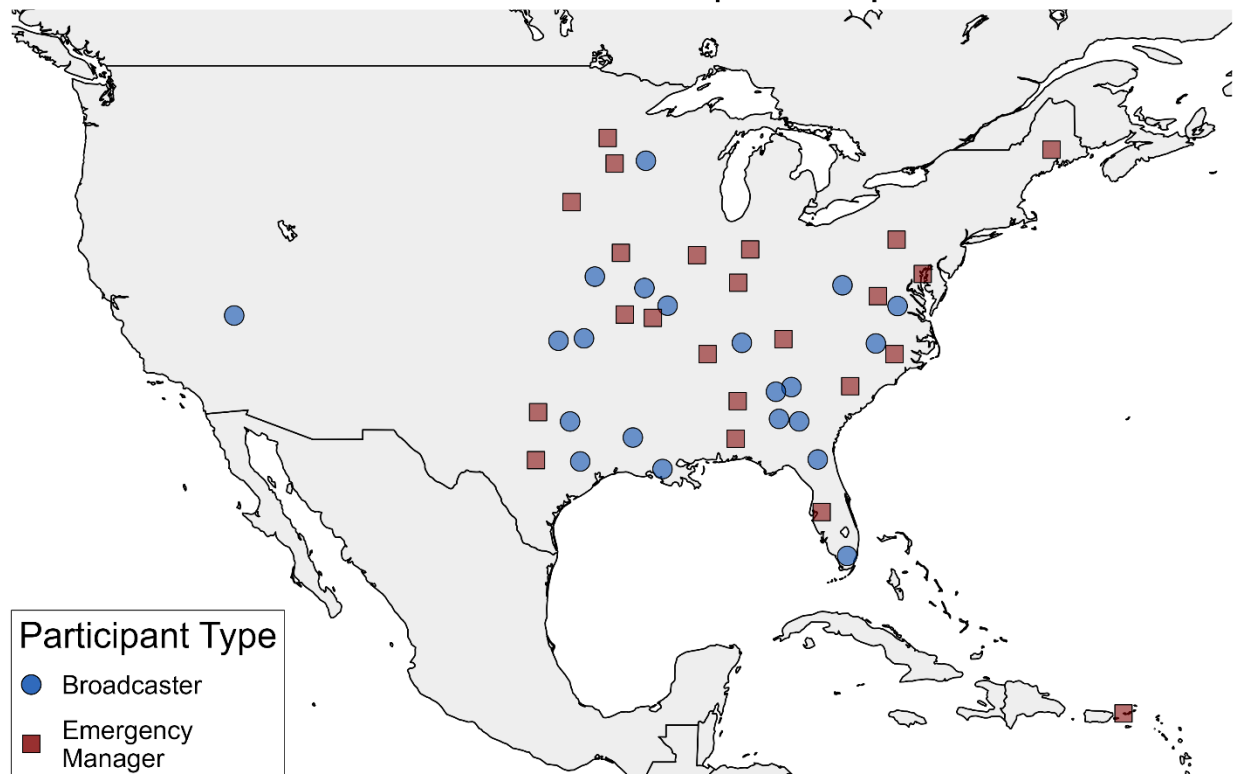


Fig. 10. Map display of the approximate locations that broadcaster and emergency manager participants in the 2024 HWT focus groups were from.

In the first of the morning sessions, moderators presented participants with the experimental categorical outlook scales and encouraged discussion around several talking points (see Appendix C for discussion guide). After being shown the four scale options, participants were asked to highlight what benefits they felt existed for each of the scale options, followed by the challenges or problems they saw for each. Once discussion began to slow, moderators would then ask participants which of the scale options they preferred to use for their own operations, and why and how they would do so. To close out the 50-minute session, moderators asked participants if changing the SPC scales would impact their news station or local jurisdiction

policies, and how those operations might change. Then, at the end of the experiment each day, moderators asked all participants to summarize their perspectives on the categorical labels, and discuss the differences in the perspectives between the broadcast meteorologists and emergency managers.

Data from the focus groups were collected in the form of recordings through the Zoom meeting service, which were initially transcribed using Adobe Premiere Pro. The automatic transcriptions were first cleaned and de-identified by three of the authors, where participants were given coded names (e.g. EM7, BCST5) that represent their participant number and their profession. We will use these coded names to reference individual participants throughout the results of this study. After de-identification, the research team performed a codebook thematic analysis (Gale et al. 2018; Braun and Clarke 2022) assisted by the coding software platform NVivo. First, two researchers built a codebook using a deductive coding scheme that would capture first-layer participant feedback about the positive and negative traits of the categorical scale options (Table 2). The transcripts were then reviewed in detail by one researcher and coded based on this deductive scheme. During the coding process, the researcher noted any themes that developed from the coded excerpts and included them in the codebook, which was then reviewed between the two researchers that developed the deductive code set. The research team also met to reconcile the final code set and organization of themes and concluded that the codes effectively captured the participants' statements and the themes were supported by the collected excerpts.

Table 2. List of deductively developed codes used to analyze focus group transcriptions, the full titles of each code, and the description of what each code was intended to capture in the dataset.

Code Title	Description
Current scale positive perceptions	Discussion of positive qualities of the current SPC outlook scale words.
Current scale negative perceptions	Discussion of negative qualities of the current SPC outlook scale words.
Likert scale positive perceptions	Discussion of positive qualities of the experimental Likert scale words.
Likert scale negative perceptions	Discussion of negative qualities of the experimental Likert scale words.

“Spanish” scale positive perceptions	Discussion of positive qualities of the experimental “Spanish” scale words.
“Spanish” scale negative perceptions	Discussion of negative qualities of the experimental “Spanish” scale words.
Numerical scale positive perceptions	Discussion of positive qualities of the experimental Numerical Level scale words.
Numerical scale negative perceptions	Discussion of negative qualities of the experimental Numerical Level scale words.
Mention of preferred or favorite scale	Participant discussion of a “favorite” or preferred scale, or that they would use one scale over another or some combination of scales.
Mention of impacts of scale changes on operations	Participant discussion of how changes to the words used in the categorical outlook would impact participant operations in their newsroom/emergency operations center.

In addition to the qualitative analysis of focus group transcripts, our EM and broadcaster participants were also given a post-questionnaire that 89% (39 of 44) of participants completed. The post-questionnaire included several questions about the different convective outlook innovations, but the relevant questions to our categorical outlook scales experiment asked participants to rank the four scales in order of preference, as well as how important they felt that it was to change the words used to convey risk in the outlook (Table 3). Note that the *Numerical* scale was referred to as the “Levels Scale” in this post-questionnaire.

Table 3. Questions related to the categorical outlook scales that were presented to participants in the post-questionnaire, as well as the response options participants had to answer each question with.

Question Text	Response Options
Please rank the four scales based on your preference. Drag and drop the options below into your order of preference:	1 - Current Scale 2 - Likert Scale 3 - “Spanish” Scale 4 - Levels Scale
How important do you think it is that SPC make changes to the words used to convey risk in the categorical outlook?	1 - Not at all important 2 - Not very much important 3 - Somewhat important 4 - Very much important 5 - Extremely important

b. Results

During both the EM and broadcaster focus groups, participants were quick to begin discussing their perspectives on the different word scales, especially with regards to the *Current* categorical outlook words. Although a small number of participants (4 broadcast, 1 EM, see Table D1 in Appendix D) mentioned that the consistency of and their familiarity with the *Current* scale was an asset, a significant majority (17 broadcast, 17 EM, out of 21 and 23 participants respectively) felt negatively about the five SPC categorical outlook words and their ordering. A majority of participants (12 broadcast, 15 EM) reported feeling like “the categories are all very subjective. What is slight to someone might be very different to what is slight to someone else” (EM7). Some of the broadcast participants noted their concerns about viewers struggling to interpret the *Current* scale, as BCST5 often “ended up getting questions about severe weather [products], whereas I personally have never gotten questions about a hurricane center product’s names and what they’re calling things.” One broadcaster even noted that, in a survey of their Facebook followers, “85% of people that didn’t know what an SPC outlook was thought ‘Enhanced’ meant something more serious than ‘Moderate’” (BCST15).

A second concern with the *Current* scale these EMs and broadcasters had was that the meanings of each of the words in the five-level scale did not match the risk that they felt each level should be communicating (10 broadcast, 4 EM). Many, like EM15, mentioned feeling that they had to translate the *Current* scale into something more intuitive for non-experts, as “usually when I am trying to translate like what a ‘Moderate’ risk is to people, [I’d say] it’s actually a really high risk. There’s a really good chance that we’re going to see severe weather in the area today.” The participants particularly feared that members of the public would dismiss a severe weather threat based on the words, because “if we tell our viewers that the chance of severe weather is low, they’re like ‘cool, gotta go.’ And they don’t continue [to seek information] beyond that point” (BCST11). In general, the broadcasters and EMs feared that even though a “level two” is a “pretty big deal... we’re assigning it [a word] so minimal, like ‘Slight’, it sends a very opposing message to the community” (EM13).

Although the *Current* scale was usually the first topic participants brought up in the focus groups, these broadcasters and EMs quickly pivoted to highlight the positive aspects of the proposed alternative scales. Participants felt that all three options - the *Likert* scale, the “*Spanish*”

scale, and the *Numerical* scale - communicated risk more effectively than the *Current* scale, though their reasons differed for each. The *Likert* scale was considered by eight broadcasters and eight EMs to present a better “spectrum” of words, particularly for the lowest levels of the scale, where “‘Very Low’ is a better way to put it than ‘Minimal’ or even ‘Marginal’” (EM11). Broadcasters also noted that “everybody is very comfortable with the ‘Low’, ‘Medium’, and ‘High’, and adding the term ‘Very’ to either end of that scale... seems to work very well” (BCST2). Similarly, some participants (10 broadcast, 4 EM) felt that the “*Spanish*” scale used “terms that people would normally use to describe things” (EM14), and would communicate risk more effectively because “I don’t think there’s a lot of room for interpretation or confusion of those words when they’re next to each other” (BCST11). Notably, the *Numerical* scale saw the broadest support, as 16 broadcasters and 12 EMs suggested that it improved on the *Current* scale by presenting a more familiar risk scale with better defined levels. Participants noted that “people are used to numbers... we do it in hurricanes, category one through five, tornadoes are zero through five” (BCST3), and that “it was very clear that everyone understood that ‘Level Two’ is more severe than ‘Level One’” (EM7). The general consensus participants reached regarding the numbers was summarized well by BCST5:

People have become trained to know what a level one day is, what a level two day is. When we go from 2 to 3... We're stairstepping it up one. [It's nice to be] able to explain it relatively quickly on air without having to get into the weeds of ‘Slight’ is less than ‘Enhanced’ (BCST5).

Despite their perceptions of the positive aspects of the three experimental scales, the participants did have some concerns about these alternative categorical outlook options. First, several participants (6 broadcast, 2 EM for both the *Likert* and “*Spanish*” scales) noted that there are words used in the *Current* scale that are used at different levels of the *Likert* and “*Spanish*” scales (see Fig. 2, specifically the words “High” and “Moderate”). These broadcasters and EMs were concerned that people might not understand that “[“High”] used to be a five out of five. Now it’s a four out of five” (BCST13). However, as this participant alluded to, eight of the broadcasters and six EMs suggested that the *Numerical* scale would be easier to implement in their operations because it does not re-use any *Current* scale words. Some participants even mentioned that they already use a number scale when presenting the categorical outlook, because “we get a lot of complications of individuals getting hung up on the category rather than getting

hung up on the risk” (EM10), or that after transitioning to numbers “everybody was a little more on the same page... the questions went away of, well... is it a ‘Slight’ or an ‘Enhanced’” (BCST5). To summarize the conclusions they had reached during the discussion, one EM noted that “the different words mean different things to different people, and they can’t be certain that [each word is] going to mean the same thing across the board” (EM9).

Another concern that participants highlighted for all four scales was the difficulty of presenting risk information for the ends of the five-level scale. For both the *Likert* scale (13 broadcast, 6 EM) and “*Spanish*” scale (6 broadcast, 3 EM), both EMs and broadcasters noted that presenting a severe weather risk as “Very Low” or “Minimal” could lead people to underestimate the risk posed to them. One broadcaster explained that “if I tell [my wife] there’s a ‘Very Low’ chance for severe weather, she is tuned out of the weather that day” (BCST1) while EM17 lamented that “if you tell someone there’s a ‘Very Low’ risk of severe weather today, then my question is why are you even raising the issue in the first place”. These concerns also existed for the word “Extreme” as used in the “*Spanish*” scale, particularly among EMs (4 broadcast, 7 EM), although some participants, more often broadcasters, found it to be a positive aspect of the scale (7 broadcast, 3 EM). Those concerned about the word “Extreme” suggested that “it’s got the fear factor in there, it’s like you’re trying to scare somebody” (EM18), or that “the word ‘Extreme’ can sometimes be taken to the extreme” (BCST12). However, those that felt “Extreme” benefited the “*Spanish*” scale noted that “an ‘Extreme’ says, okay, batten down the hatches, we’re going to get crushed” (BCST10) and that “sometimes ‘Extreme’ may scare people... [but] if they’re not scared, they rarely take it seriously” (EM22). Finally, several participants (3 broadcast, 4 EM) also mentioned that they were concerned that presenting the lower level numbers may also result in people underestimating severe weather risks, especially when in some areas “moderate may actually be the highest we ever see... so I wouldn’t want to put a level to it because it still could be at least significant based on local factors” (EM2).

In addition to their concerns about the worded scales, a number of participants (8 broadcast, 7 EM) worried specifically for the *Numerical* scale that completely removing words from the categorical outlook would negatively impact risk communication. They felt that numbers fail to communicate the emotional valence of the different risk levels and could be misinterpreted by members of the public. One broadcaster summarized these feelings well in a theoretical example:

“Three out of Five”, it’s clear, it’s concise. But “Three out of Five”, does mom at home getting her three-year-old kid ready for school or getting ready to walk out the door to work... [she may not be] sure what that means. So I think it can’t just be numbers, but it’s got to be a clear word to be able to catch somebody’s attention (BCST17).

Emergency managers shared similar concerns, as they suggested that “it’s a little harder for people to wrap their heads around a number, because ‘One out of Five’... What does that mean? Five meaning it’s absolutely going to happen. Or five meaning it’s very likely to happen. What does that five represent” (EM14). Participants in general noted that the inclusion of a word in the categorical outlook made their communication efforts easier, because “it’s nice to have the word... we can further elaborate by saying the number to kind of make things a little bit easier to understand” (BCST9).

Finally, when asked what impact changing the categorical outlook scale words would have on their severe weather operations, the majority of participants (18 BCST, 16 EM) suggested the change would have little to no impact. Most EMs noted that the change would only involve “slight wording changes in the [standard operating procedure]” (EM5), because “what the level is called doesn’t dictate what we do. It’s the level itself that dictates what we do” (EM14). For broadcasters, “these levels are all going to be calculated the same way... like the 15% tornado does the same thing now... for a news operation it would just be a little hey, this is the new thing and this is the way it is now” (BCST13). The only concern participants voiced about transitioning from the *Current* scale to a new alternative was that they might need to make additional efforts to educate others on how to interpret the new scale (5 BCST, 8 EM). EMs in particular highlighted how “there is a little bit of education we would need on our end to tell our elected officials exactly what these changes mean” (EM2), while for the Spanish and Likert scales broadcasters noted that “the only thing I would say would be the re-education part on the part of the entire newsroom” (BCST15). The general belief that impacts of a scale change would be minimal was reflected in the post-survey results, as well, with the majority of participants suggesting the change would impact their operations “not very much” or “somewhat” (see Fig. 11a).

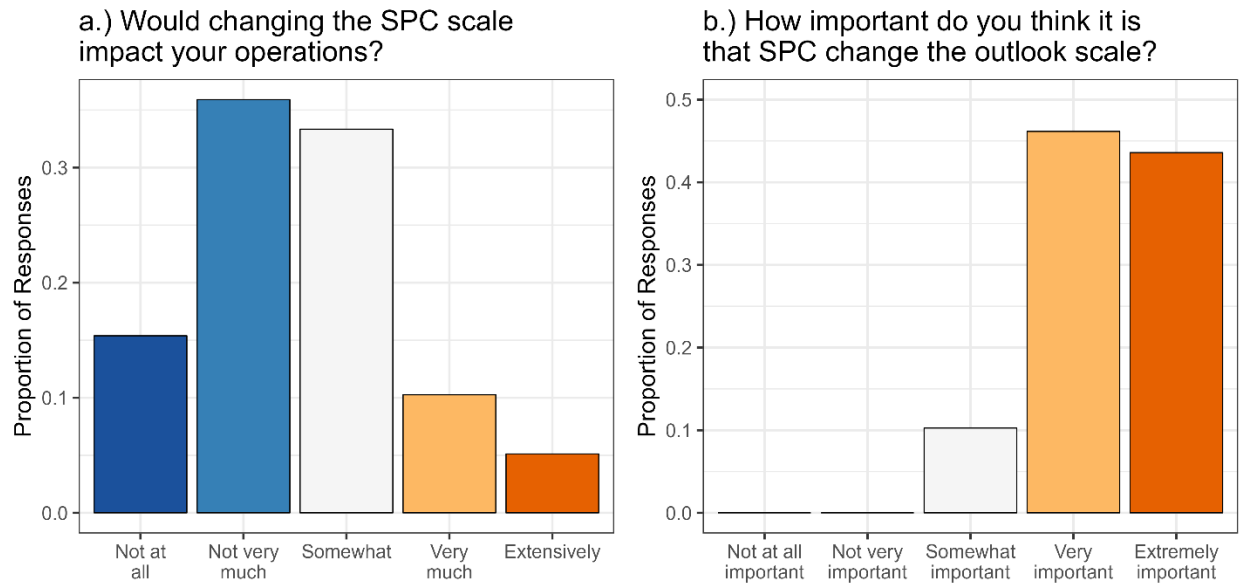


Fig. 11a-b. Proportions of responses to the focus group post-questionnaire questions a.), “How much would a change to the word scales for the Convective Outlook affect your operations and organizational policies for severe weather events” and b.), “How important do you think it is that SPC make changes to the words used to convey risk in the categorical outlook?” Responses were collected using a 5-point scale, which ranged from Not at all important to Extremely important. When these levels are converted to numbers ranging from one to five, the mean response was 2.54 for a.), between Not very much and Somewhat, and 4.33 for b.), between Very and Extremely important.

To conclude the focus groups, participants were also asked to highlight their preferences across the different scales. While many chose one particular scale as their favorite, many others mentioned that they liked two of the scale options about equally (see Table 3). Overall, the *Numerical* scale was most popular with the broadcaster participants and participants overall, while EMs preferred the “*Spanish*” scale. Combining the *Numerical* scale with the “*Spanish*” scale words was also the most frequently proposed “ideal” scale, mostly due to of its popularity with broadcasters (see Table 4). Similar results were also found in the post-survey when participants were asked to rank the four scale options, where the *Numerical* scale and “*Spanish*” scale were most popular, followed by the *Likert* and *Current* scale (see Fig. 12a). Notably, broadcasters overall strongly endorsed the *Numerical* scale (see Fig. 12b), while EMs scale preferences varied much more for all four options (see Fig. 12c). However, all participants agreed that changing the categorical outlook words was at least somewhat important, with an

average response on the 5-point scale of 4.33 (between Very and Extremely Important, see Fig. 11b) in the post-survey.

Table 4. Count of participants that, when asked in the focus group to state their preference for which scale option they would prefer to use, expressed their preference each scale or combination of scales for the categorical outlook. The highest number of respondents for each scale preference is bolded in the table for ease of reference.

Stated Scale Preference	Broadcaster Count (Percentage)	Emergency Manager Count (Percentage)	Total (Percentage)
Current Scale	0	0	0
Likert Scale	0	4 (17.4%)	4 (9.0%)
“Spanish” Scale	2 (9.5%)	6 (26.0%)	8 (18.2%)
Numerical Scale	6 (28.6%)	4 (17.4%)	10 (22.7%)
Numerical OR Likert	3 (14.3%)	2 (8.7%)	5 (11.4%)
Numerical OR Spanish	5 (23.8%)	0	5 (11.4%)
“Spanish” OR Likert	2 (9.5%)	4 (17.4%)	6 (13.6%)
No Preference Mentioned	3 (14.3%)	3 (13.0%)	6 (13.6%)

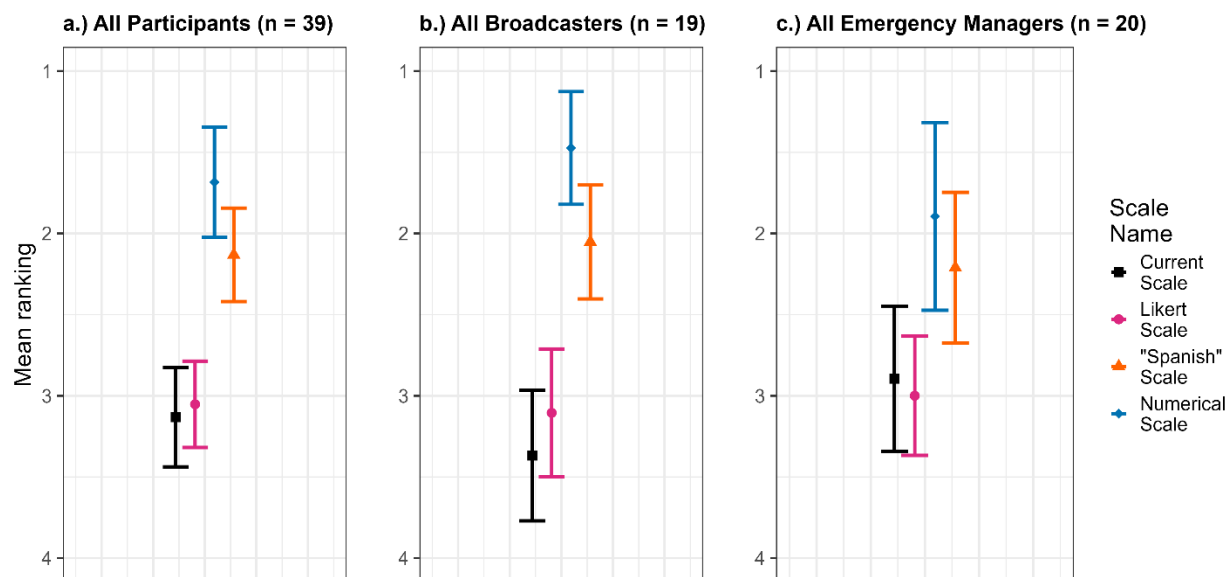


Fig. 12a-c. The mean ranking of the four scale options tested in the focus group post-questionnaire, for all respondents in a), broadcasters in b), and EMs in c). Error bars represent a 95% confidence interval based on standard error.

5. Discussion

a.) Comparing findings to hypotheses

At the beginning of this study, we hypothesized based on prior investigations into core partner and public interpretation of the categorical outlook scale and alternatives to it (Ernst et al. 2021; Williams et al. 2022; Krocak et al. 2022; Bitterman et al. 2023a), that 1) the *Numerical* scale would see more linear increases in average perceived concern and likelihood of protective action than the other scales, with the *Current* scale the worst performer; and 2) that perceived response efficacy would be consistent across the *Numerical* scale but decrease for the highest levels of the word scales. With our survey experiment, we found that the first hypothesis was supported by the data, as average participant concern and likelihood of response increased most linearly across the *Numerical* scale, somewhat linearly for the *Likert* and “*Spanish*” scales, and not linearly for the *Current* scale (see Figs. 4, 6). These observed patterns changed significantly, however, when participants were asked the same questions after being shown a forecast map that presented a categorical outlook, their location within it, and a legend with all the scale words and colors included on it (see Fig. 3a-f). Participants displayed a generally linear increase in concern and likelihood of protective action across all of the categorical scales (see Fig. 7), suggesting that the contextual information provided by the map and the legend led users to calibrate their concern to what the scale is attempting to communicate.

Our survey also helped us test our second hypothesis, which we rejected as we identified that participants’ perceptions of whether actions they could take to protect themselves from severe weather hazards would be effective, or response efficacy, did not vary across the different categorical outlook word scales. Instead, response efficacy was consistent and high across all levels of each scale despite the increasingly strong language being used. This finding presents an interesting comparison with Ripberger et al. (2015), as unlike in that study we did not ask participants what protective action they thought would be efficacious for the different risk levels presented in our survey. However, we do not see a decrease in response efficacy for the higher-

level words, suggesting that individuals were not questioning their ability to keep themselves safe when higher end threats are forecast for them.

In addition to our public survey data analysis, we explored core partners' preferences for the four categorical outlook scale options in a series of focus groups and questionnaires. In these focus groups, emergency managers and broadcasters expressed feeling that the *Current* scale ("Marginal", "Slight", "Enhanced", "Moderate", "High") was difficult to understand because the words are too subjective in their meaning and do not match the severe weather risk that each level is meant to communicate. The practitioners we spoke to generally preferred the *Numerical* scale, although some participants feared that a solely number-based scale would not communicate the emotional weight that they felt words, like those in the "*Spanish*" scale, are able to. Importantly, the focus group participants did not think that changing the categorical outlook words would have a significant negative impact on their operations and strongly supported moving away from the *Current* scale.

Given the results of our two studies here, in addition to previous studies of the categorical outlook (Ernst et al. 2021; Krocak et al. 2021; Williams et al. 2022; Krocak et al. 2022; Bitterman et al. 2023a), the categorical scale most preferred by NWS partners with the clearest communication to members of the public of a linear increase in risk across it is the *Numerical* scale. Although not a perfect communication tool, with some small disruptions in the linearity of the increase in mean public concern and likelihood of response noted where the differences between the confidence intervals overlap with the means for each level, the *Numerical* scale performs well when presented alone or with a map (see Figs. 4-9) while the other scales exhibited smaller changes or even decreases in the means between levels when only the level was shown to participants. It is important to recognize that even though all scales saw more linear increases in risk when maps and legends were presented to participants, there are many information channels where only textual or limited visual risk information is communicated that are still frequently used, including social media and radio (Bitterman et al. 2023b). Thus, we agree with Krocak et al. (2022) that a solely *Numerical* scale is the clearest, most intuitive way to communicate risk in the categorical outlook in every format the outlook could potentially be encountered in.

The *Numerical* scale was also the preferred choice among the four scales for the focus group participants, who also agreed that changing from the *Current* categorical risk words was

important to do. However, the emergency manager participants most often chose the “*Spanish*” scale as their favorite in the focus groups and on average ranked the *Numerical* and “*Spanish*” scales similarly highly in the post-questionnaire. Some of the concern that focus group participants voiced about the effectiveness of the *Numerical* scale likely stems from their desire for a scale to include words that accompany the numbers for each level, which we identified is a common practice with other number-based risk scales (Fujita 1971; WSEC 2006; NCEI 2025; SWPC 2025; EHP 2025). Further, for many of the words in the *Current* scale and the words at the high and low ends of the *Likert* and “*Spanish*” scales, focus group participants emphasized *feeling* that the words did not match the level of seriousness that they wanted those levels of the categorical outlook to communicate to non-experts.

In this case, both the practitioners and risk communicators are expressing their awareness of the affective element of risk communication. Affect, or the instant, emotional response we feel towards stimuli, has been shown to be an important part of decision-making under risk (Slovic et al. 2005; Keller et al. 2006; Terpstra 2011; Ernst et al. 2025), and that fear-based appeals can lead to increased protective action uptake (Terpstra 2011; Paek et al. 2016; Ernst et al. 2025) but may result in individuals choosing protective actions or coping mechanisms that are less suited to the threat at hand (Slovic et al. 2005; Lindell and Perry 2012; Ripberger et al. 2015). The broadcasters and EMs here are concerned that words like “Minimal” and “Very Low”, along with numerical levels in general, do not provide enough of a fear appeal to encourage members of the public to monitor weather conditions and expect potentially impactful storms. At the same time, they feel that words like “Extreme” may lead to an excessive fear response that could lead people to pursue coping strategies that could put them in danger. However, these concerns contrast with our finding that members of the public’s mean response efficacy was consistent across or highest at the highest scale levels for all four scales we tested in our survey, suggesting participants felt that their protective actions of choice would effectively reduce their risk from severe weather even when told their risk was “Extreme”. Combined, these results suggest that the relative unfamiliarity of a *Numerical* categorical outlook scale, especially given the use of words in the *Current* scale, is the primary driver for core partner hesitation about transitioning to a fully numbers-based scale.

The hesitation expressed by some forecast users towards implementing a numbers-based risk scale does suggest that some aspects of the word-based scales should be carried over to a

new scale design, and recent research evaluating similar risk communication scales further supports this idea. First, given prior reviews of risk scales have identified the value of incorporation of user feedback into scale design (Fearnley et al. 2012; Potter et al. 2014; Fearnley and Beaven 2018; Semmens et al. 2023, 2024; Kastman et al. 2025), the concerns expressed by EMs and broadcasters about the need for words that communicate affect suggest that any Numerical scale should be accompanied by a legend that includes keywords that describe the severe weather risk each level is associated with. As for what these keywords should be, the focus group participants (in particular the EMs, see Fig. 12c) favored the “*Spanish*” scale words over the *Likert* and *Current* words, which also saw more consistent increases in likelihood of response across all levels than the other word scales (see Figs. 5, 8). The “*Spanish*” scale also uses the English translations of a thoroughly tested and now well-established set of Spanish dialect-neutral risk communication words (Trujillo-Falcón et al. 2021, Bitterman et al. 2023a), meaning that translation efforts between the English and Spanish uses of the “*Spanish*” scale would be simplified. Including the “*Spanish*” scale words in a legend that describes the Numerical scale might also avoid issues where forecast users seek greater contextual information about potential hazard impacts, as was identified for the Drought Index by Ward et al. (2022).

b.) Limitations

Although this study was intended to summarize the work done to understand categorical outlook interpretation and advance that research to the point where a recommendation on proper practice for presenting severe weather hazards with a verbal scale could be made, this work was not without its limitations. Due to the design of the public survey, the 1513 participants overall were randomly broken into groups of roughly 75 participants, each of which were shown one of the 20 words included in the four scale options. The small sizes of these groups preclude any demographic analyses of these data, and future studies are encouraged to present word options to larger groups of participants in surveys or to specific demographics in focus groups to learn more about how they react to these scale options. Additionally, although we included a large number of emergency managers and broadcasters in our study, these participants were reached through the HWT recruitment process, which interested participants must apply to. Thus, these participants self-selected into this study and may be more engaged with weather topics, which may not represent both groups on average. Future work should seek to reach out to broadcasters

and emergency managers that may not be as engaged in severe weather research, to gauge their perceptions of the alternatives to the categorical outlook scale. Further research into the public and partner response to changing the categorical outlook words should also be performed even if the recommendations of this study are followed, first to confirm that the positive impacts on risk perception and protective action behavior found here are replicated, but also to observe if any unexpected changes to both occur after implementation. Finally, this study focused on severe thunderstorm hazard communication, and the results of our studies may not be applicable or comparable to how message recipients respond to categorical scales for other hazards. Future work should identify whether consistent and inconsistent elements across scales for different hazards help or hinder the interpretation of those scales, especially when multiple hazards may be forecast at one place and time.

6. Conclusion

When considering our results and the prior literature on SPC categorical outlook interpretation, we believe that we can recommend the *Numerical* scale as a more intuitive way to present verbal risk categories than the *Current* scale used in the categorical outlook. We further recommend that the *Numerical* scale be presented on its own in the categorical outlook (see Fig. 13), as previous work has identified that the positive benefits of using a number scale can be blunted by the inclusion of scale words (Krocak et al. 2022). We also recommend that risk communicators emphasize communication channels that allow for visual presentations of the categorical outlook, as individuals interpret the product more consistently regardless of the words used for each category. Other supporting evidence for using the *Numerical* scale for verbal communication of the categorical outlook can be seen in its adoption by some NWS offices, local broadcasters, and emergency managers, who have not reported significant pushback from forecast recipients over the change to the outlook levels. However, based on the feedback we collected in our focus groups and our study of other risk communication scales, we also recommend that any application of the *Numerical* scale to the categorical outlook should be accompanied by a detailed legend that uses the user-preferred “*Spanish*” words to contextualize each level of risk in a summary for each level (see Fig. 14). Providing categorical words in the descriptions of each level in the scale legend may address the concerns some practitioners have shared in our study and others (Trujillo-Falcón et al. 2024; Gaviria Pabón et al. 2025) about how

the *Numerical* scale would not be intuitive and would not communicate risk in a way that would encourage instant affective responses. Precedent for categorical word use in the legend of hazard risk scales also exists, as operational scales including the EF-scale, the Saffir-Simpson scale, and even the MMI earthquake shaking scale also use words, like “minor” or “catastrophic”, to describe the potential or observed impacts of the hazards they capture. We recommend that the legend descriptions should use the categorical words in the “*Spanish*” scale because of their similarity to the words used in those operational risk scale legends, in addition to broadcaster and emergency manager’s expressed preference for that word scale over the others. Further, the “*Spanish*” scale’s pre-existing translation would help bilingual broadcasters continue to add desired contextual information (Trujillo-Falcón et al. 2024; Gaviria Pabón et al. 2025) when presenting a numerical categorical outlook without additional development efforts. In summary, we suggest that by combining the *Numerical* scale with a detailed legend, as in the NWS HeatRisk product (NWS 2025), the categorical outlook will be able to more clearly and intuitively present easily translated levels of severe weather risk to both core partners and the public.

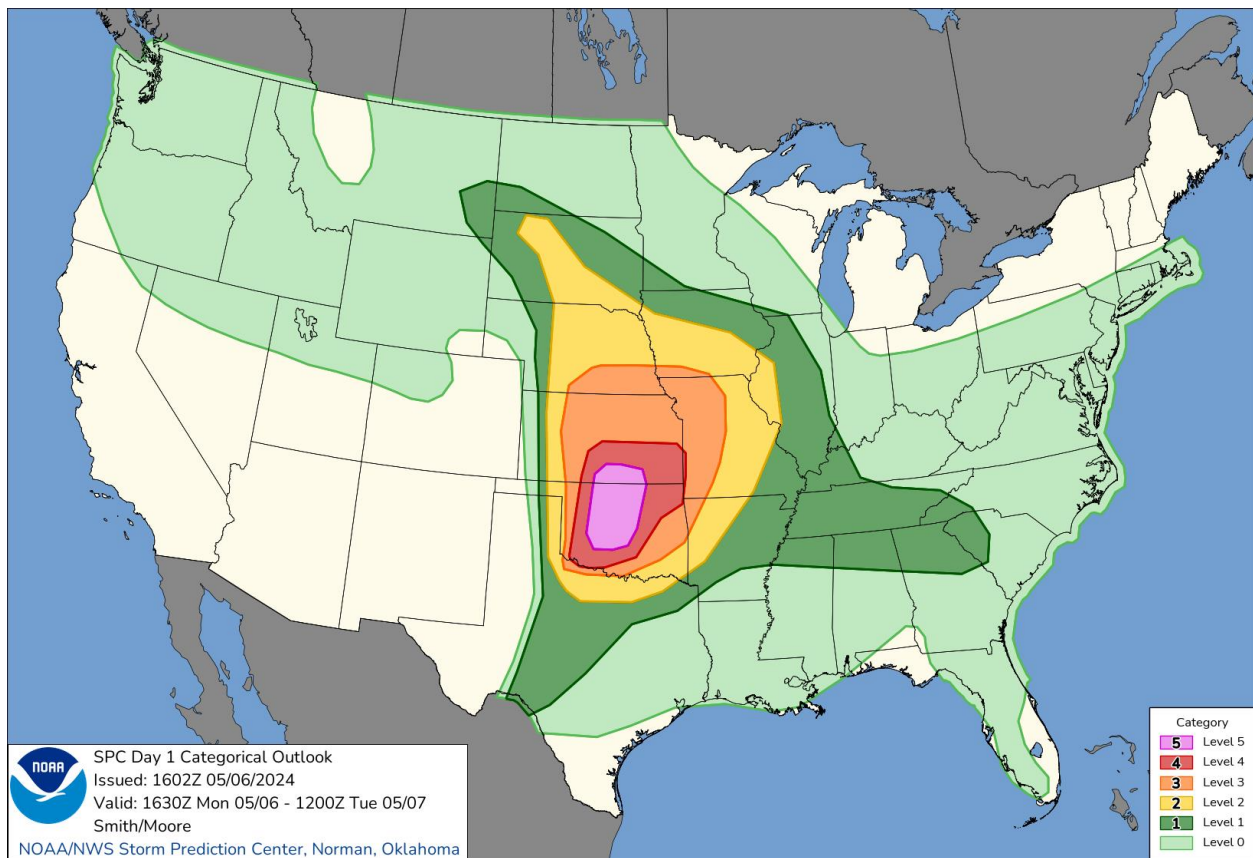


Fig. 13. Example of how the categorical outlook could be visually presented, using the *Numerical* scale in the place of the *Current* words. This presentation is based off of the same event as Fig. 1, for comparison purposes. Image created by Nathan Wendt and used with permission from the SPC.

Level	Summary	Probability of Hazards within 25 miles	How many severe storms are possible?
Level 0	General Thunderstorms are possible in this region, but severe thunderstorms are <i>not expected</i> . Be aware that all thunderstorms can produce deadly lightning, gusty winds, small hail, and flooding rains.	0% tornado risk 0% hail risk 0% wind risk	None expected.
Level 1	Minimal risk of a <i>few</i> severe thunderstorms, which <i>could be capable</i> of damaging winds and/or severe hail. Tornadoes are unlikely, but stronger storms could pose a tornado threat.	2% tornado risk 5% hail risk 5% wind risk	Isolated severe storms are expected.
Level 2	Low risk of several severe thunderstorms, with <i>medium confidence</i> that <i>several</i> storms will have damaging winds, severe hail, and/or tornado potential. There is a <i>low likelihood</i> that a several storms will be significantly severe.	5% tornado risk 15% hail risk 15% wind risk	Isolated to scattered severe storms are expected.
Level 3	Moderate risk of several severe thunderstorms, with <i>high confidence</i> that <i>several</i> storms will have damaging winds, severe hail, and/or tornadoes. There is a <i>medium likelihood</i> that several storms will be significantly severe.	10-15% tornado risk 30-45% hail risk 30-45% wind risk	Scattered to numerous severe storms are expected, with a <i>low chance</i> of a <i>severe weather</i> outbreak.
Level 4	High risk of severe thunderstorms, with <i>high confidence</i> that <i>many</i> severe thunderstorms will have damaging winds, severe hail, and/or tornadoes. There is a <i>high likelihood</i> that several storms will be significantly severe.	15-30% tornado risk 45-60% hail risk 45-60% wind risk	Numerous severe storms are expected, with a <i>medium chance</i> of a <i>severe weather</i> outbreak.
Level 5	Extreme risk of an <i>outbreak of severe thunderstorms</i> , which will have damaging winds, severe hail, and/or tornadoes. <i>Many</i> significantly severe storms are <i>expected</i> , posing a risk for strong tornadoes and hurricane-force winds.	>30% tornado risk 45-60% hail risk 45-60% wind risk	Very numerous severe storms are expected, with a <i>high chance</i> of a <i>severe weather</i> outbreak.
Note: Level names, Summary content, and description of how many severe storms are possible was developed through interdisciplinary meteorology and social science research, published in Trujillo-Falcón et al. 2021; Ripberger et al. 2022; Krocak et al. 2022; Bitterman et al. 2023; and Ernst et al. 2026.			

Fig. 14. Example of how a legend for the numerical scale for the categorical outlook could be presented, using the “Spanish” scale words to help summarize what hazards each level of the Numerical scale is communicating. This legend design is based on a combination of the findings of this study, the recommendations of Ripberger et al. (2022) on verbal probabilistic communication, and the current SPC outlook legend.

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Data Availability Statement.

Survey data analyzed for this experiment is available online at the Harvard Dataverse, and can be found at the Harvard Dataverse at the following link:

<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/PPXI8H>. The code used to analyze survey data and a de-identified version of the post-questionnaire data can be found online at

<https://github.com/ippra/wxsurvey/tree/master/offering%20an%20evidence%20based%20recom%20mendation%20for%20improving%20the%20storm%20prediction%20centers%20categorical%20outlook%20with%20a%20numerical%20scale>. Qualitative data collected during focus groups can be obtained in a de-identified format via email from the corresponding author.

APPENDIX

Appendix A: WX23 Demographic Benchmarks

Table A1: Demographic margins of the U.S. Adult Population as per the 2020 US Census vs WX23 respondent percentages, for demographic variables used in the weighting of survey data.

Demographic Variables	Census Benchmarks	Survey Sample (Unweighted)
Age		
18 to 29	20.00%	18.52%
30 to 49	33.60%	35.49%
50 to 64	23.80%	24.26%
65 and up	22.60%	21.72%
Education		
High School or Less	37.50%	25.27%
Some college/2-year degree	28.90%	34.96%
4-year/post-graduate degree	33.60%	39.77%
Gender		
Female	49%	46.59%
Male	51%	53.41%
Income Group		

< \$50,000/year	27.60%	48.26%
> \$50,000/year	72.40%	51.74%
Race		
White	60.10%	67.91%
Black or African American	11.70%	11.23%
Hispanic	17.60%	15.57%
Other Race	10.60%	5.28%
NWS Region		
Eastern	31.40%	31.28%
Southern	27.70%	27.07%
Central	20.40%	20.52%
Western	20.50%	21.12%

Appendix B: Unweighted WX23 Survey Question Results

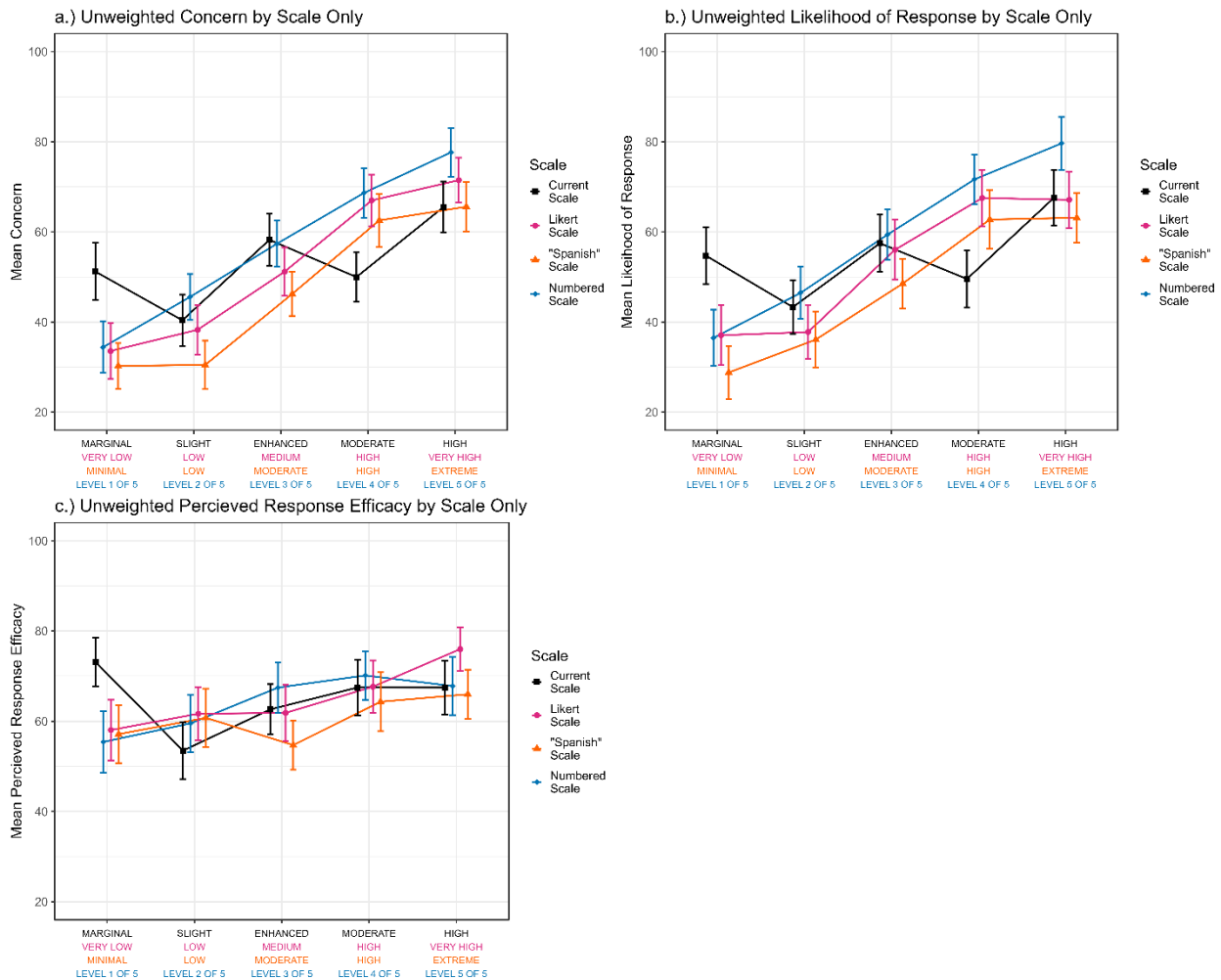


Fig. B1a-c. Plots equivalent to Figs. 4-6 but instead generated using the unweighted data from WX23, where participants were shown only the risk word they were assigned. Data points represent the mean responses for level of concern in a), likelihood of response in b), and response efficacy in c), all on a 0-100 scale. Labels on the X-axis are color coded to match the scale that they are from. Error bars around each mean represent the 95% confidence interval around each, based on the standard error.

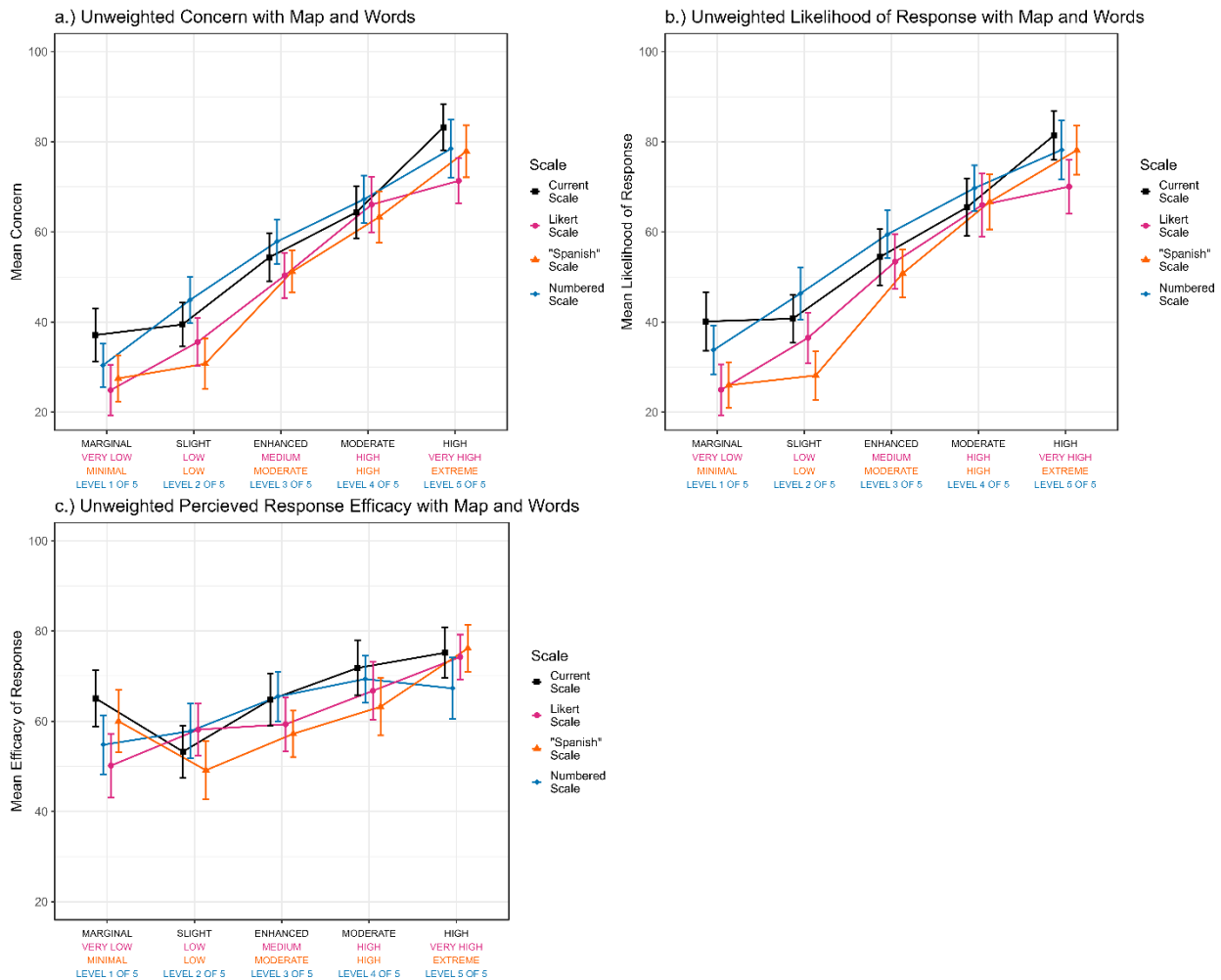


Fig. B2a-c. Plots equivalent to Figs. 7-9 but instead generated using the unweighted data from WX23, where participants were shown their location on a map with a legend depicting the five levels of their assigned categorical outlook scale. Data points represent the mean responses for level of concern in a), likelihood of response in b), and response efficacy in c), all on a 0-100 scale. Labels on the X-axis are color coded to match the scale that they are from. Error bars around each mean represent the 95% confidence interval around each, based on the standard error.

Appendix C: Focus Group discussion prompt

Alternative SPC Word Scale Discussion - Broadcast Meteorologists

*Note for the leader: It is your job to **moderate** this focus group, **NOT** to contribute. You are to ask prompting questions to lead the participants to discuss, **NOT** to lead their responses. It is appropriate to repeat what someone has said for the group and ask for clarification if unclear. It is appropriate to ask “how”, “what”, or “why” questions to get participants to follow up and round out their ideas. Do not offer your own examples unless directly asked. Do not offer your thoughts on the outlook.*

(9:15-10:05) Leader:

Now that we are starting our focus groups, I am going to begin recording, and remind our note-takers to do the same!

BEGIN RECORDING SESSION

For our first focus group today, we’re going to discuss the first innovation the SPC is looking at applying to the convective outlook - namely, a change to the English words used to describe each level of the categorical outlook.

As a reminder, this would not change the convective outlook as it currently stands, just the words used to describe each level. The outlook will remain a 5-level scale with a probabilistic forecast describing what each level is for.

Pull up single slide/page with the four scales written out so that non-audial learners or those hard of hearing can read along

The SPC has been testing **four** separate versions of the SPC outlook:

1. The **Current Scale**, which is “**Marginal, Slight, Enhanced, Moderate, and High**”.
2. The **Likert Scale**, which is based on a common organization used in social science. This scale is “**Very Low, Low, Medium, High, Very High**”.

3. The “**Spanish**” **Scale**, which is the direct English translation of the recently adopted and dialect-friendly Spanish categorical outlook scale. This scale is “**Minimal, Low, Moderate, High, Extreme**”.
4. And finally the **Levels Scale**, which uses numerical levels like those used for recording tornado and hurricane intensity. This scale is “**Level 1 of 5, Level 2 of 5, Level 3 of 5, Level 4 of 5, and Level 5 of 5**”.

We want to know what you, as the key communicators of weather forecast information to the public, think of these different options for communicating convective weather risk 1 - 8 days in advance.

Minimize screen. Pause. Allow participants to fill empty space. If too silent, move to the following questions.

1. What benefits do you feel exist for each of these scale options?

*Pause. Allow participants to fill empty space. Have the **leader** or **notetaker** PASTE the four options written above into the chat so that participants can refer back to options easily during conversation.*

2. What challenges or problems do you see with each of these scale options?

Pause. Allow participants to fill empty space. Guide through each four to make sure every option is discussed by participants.

3. Which of these scale options would you prefer to use for your own operations?

- a. **Why?**
- b. **How?**

Pause. Allow participants to fill empty space. Guide through each four to make sure every option is discussed by participants.

- 4. Would changing the SPC outlook levels from the current scale to one of these scale options impact your station's response policies?**
 - a. How?**

Pause. Allow participants to fill empty space. Guide through each four to make sure every option is discussed by participants.

(10:05-10:30) Leader:

We have also studied these scales with members of the public in a few different surveys. We are going to show you a few results from that survey.

Present slides about the public response to scales. (5 mins!)

- 1. Now that you have seen the public's responses to these proposed scales - has your perception of these scales changed?**
 - a. How?**

Pause. Allow participants to fill empty space.

- 2. Has this changed your scale preference?**

Pause. Allow participants to fill empty space.

Appendix D: Table of theme mention counts

Table B1: Table of themes with the number of participants that were coded as presenting that theme during the focus groups. Participants are broken down into broadcasters and emergency managers (EMs).

Theme	Broadcaster Mentions (out of 21)	EM Mentions (out of 23)
Current scale is familiar to users	4	1
Current scale uses overly subjective words	12	15
Current scale words do not match severity of hazards	10	4
Likert scale has a better spectrum of words	8	8
“Spanish” scale uses words people actually use	10	4
Number scales are familiar to people	16	12
Terms in the Likert scale overlap with the Current scale	6	2
Terms in the “Spanish” scale overlap with the Current scale	6	2
The Numerical scale is easier to implement with no words	8	6
The low end of the Likert scale is dismissive of risk	13	6
The low end of the “Spanish” scale is dismissive of risk	6	3
The high end of the “Spanish” scale is too alarming	4	7
The high end of the “Spanish” scale is effective	7	3
The low end of the Numerical scale is dismissive of risk	3	4
The Numerical scale needs words to communicate emotion	8	7

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